MATHEMATICS GRADE 10: ROTATION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.2 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 3.0: Geometry
Sub-Strand	Sub-Strand 3.2: Rotation
Total Duration	10 lessons (approximately 20 periods × 40 minutes = 800 minutes)
Sub-Strand Content	– Properties of rotation – Angle of rotation – Centre of rotation – Rotating points and shapes on the Cartesian plane – Determining the angle and centre of rotation given an object and its image – Order of rotational symmetry – Axis of rotational symmetry – Congruence and rotation – Applications of rotation in real-life situations
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) identify and describe the properties of rotation in different situations, b) rotate points and shapes through a given angle about a given centre on the Cartesian plane, c) determine the angle and centre of rotation given an object and its image, d) determine the order and axis of rotational symmetry of plane figures, e) relate rotation to congruence of shapes, f) apply rotation to solve problems in real-life situations, g) appreciate the use of rotation in everyday contexts such as design, engineering, and nature.
Core Competencies	– Communication and Collaboration: the learner discusses with peers to identify and describe the properties of rotation and shares findings on rotating shapes on the Cartesian plane. – Critical Thinking and Problem Solving: the learner determines the angle and centre of rotation given an object and its image, and applies rotation to solve real-life mathematical problems. – Creativity and Imagination: the learner designs patterns and figures using rotational symmetry, exploring applications in art, architecture, and nature found in Kenya. – Digital Literacy: the learner uses digital devices and simulation tools to explore rotation of points and shapes and to verify results obtained by hand construction.
Core Values	– Unity: the learner collaborates in group activities to explore the properties of rotation and determine centres and angles of

	rotation. – Responsibility: the learner takes care of mathematical instruments (protractors, compasses, rulers) and digital devices used during rotation constructions. – Respect: the learner listens to and values contributions from peers during group discussions and presentations on rotational symmetry. – Integrity: the learner accurately records and reports measurements and constructions without falsifying results.
Science & Engineering Practices	– Developing and Using Models: the learner builds and revises cumulative diagrams/models on the Cartesian plane to represent how a shape transforms under rotation, tracking changes in position and orientation. – Analyzing and Interpreting Data: the learner measures and records angles and coordinates of object and image points, then interprets patterns to determine the centre and angle of rotation. – Constructing Explanations: the learner uses observed properties of rotation to explain why object and image are always congruent and why distance from the centre of rotation is invariant. – Engaging in Argument from Evidence: the learner justifies the identified order of rotational symmetry of a figure by providing symmetry evidence from multiple orientations. – Obtaining, Evaluating, and Communicating Information: the learner reads, evaluates, and communicates findings from video resources, simulations, and readings about rotation in real-life contexts.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Critical Thinking): the learner applies logical reasoning to determine the angle and centre of rotation given an object and its image, building problem-solving confidence. – Life Skills (Self-Esteem): the learner participates actively in discussions and presentations on rotational symmetry, developing confidence in mathematical communication. – Environmental Education: the learner identifies rotational symmetry in natural Kenyan environments — e.g., the symmetry of flowers, Maasai beadwork patterns, and the rotating blades of wind turbines at Ngong Hills. – Safety: the learner uses mathematical instruments carefully during constructions to avoid injury and damage.
Career Connections	– Architecture and Construction: rotation is used in designing rotationally symmetric structures such as the domes of Kenyan mosques, the KICC building in Nairobi, and circular stadium layouts. – Engineering: mechanical engineers apply rotation when designing rotating parts such as turbine blades at the Ngong Hills Wind Farm and gear systems in machinery. – Graphic Design and Art: designers use rotational symmetry to create Kenyan fabric patterns, logos, and decorative motifs found in Lamu and Maasai crafts. – Astronomy and Navigation: understanding rotation underpins the study of Earth's rotation, satellite orbits, and navigation systems used in Kenyan aviation and maritime transport on Lake Victoria. – Computer Animation and Gaming: rotation transformations are fundamental algorithms used in software to animate characters and objects in Kenyan-developed mobile apps and games.
Focus for Lessons	This unit develops learners' understanding of rotation as a geometric transformation, moving from the fundamental properties of rotation to practical applications in real-life Kenyan contexts. Learners explore how shapes rotate about a fixed centre, investigate rotational symmetry in natural and man-made objects, and connect rotation to the concept of congruence. The unit uses a phenomenon-driven, model-building approach where learners progressively revise their understanding through prediction, observation, and explanation phases anchored to a compelling real-world anchoring phenomenon.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate — and how can we use the mathematics of rotation to predict exactly where any point on the blade will land?

	KICD KEY INQUIRY QUESTIONS: 1. What are the properties of rotation as a geometric transformation? 2. How do we determine the centre and angle of rotation given an object and its image?
Anchoring Phenomenon	Students are shown a video of the Ngong Hills Wind Farm turbines rotating. The blades spin continuously about a fixed central hub — each blade sweeps through equal angles in equal time, always returning to positions that look identical to the original. Students are asked: "Why does each blade always look the same no matter which position it stops at? How does the whole turbine blade assembly map perfectly onto itself?" This is puzzling because students can see that the blades move significantly through space, yet the overall shape appears unchanged — raising questions about what stays the same and what changes during a rotation, where the fixed point is, and why three blades spaced equally apart produce this effect. This phenomenon anchors the entire unit, with each lesson resolving a piece of the puzzle about how rotation works, how we measure it, and what properties are preserved.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Introduction to Rotation: Students watch the Ngong Hills wind turbine video and make initial predictions about what stays the same and what changes. Class co-constructs the Driving Question Board (DQB) with initial questions and ideas. Students read an introductory passage on rotation, establishing the need-to-know questions that will drive the unit. Lesson 2 — Properties of Rotation: Students explore the defining properties of rotation (fixed centre, invariant distances, direction of turn, angle of rotation) using cut-out shapes and protractors. They rotate shapes manually and observe that object and image are always congruent and that every point maintains its distance from the centre. Findings are connected back to the turbine blades phenomenon. Lesson 3 — Rotating Points on the Cartesian Plane: Students learn to rotate individual points by 90°, 180°, and 270° (clockwise and anticlockwise) about the origin on the Cartesian plane. They use coordinate rules and verify using a video simulation. Students update their DQB models with coordinate evidence. Lesson 4 — Rotating Shapes on the Cartesian Plane: Students extend point rotation to full shapes (triangles, quadrilaterals) on the Cartesian plane, rotating about the origin and other given centres. They construct images using protractors, rulers, and squared paper, then verify with digital tools. Lesson 5 — Angle and Centre of Rotation (Given Object and Image): Students are given object-image pairs and must determine the angle and centre of rotation by construction (perpendicular bisectors of corresponding point segments). A reading on logical reasoning in geometry supports sensemaking. Students revisit the turbine: "If we know one blade position and its rotated image, can we find the centre?" Lesson 6 — Order of Rotational Symmetry: Students investigate which regular shapes (equilateral triangle, square, regular hexagon, etc.) map onto themselves during a rotation of less than 360°. They define order of rotational symmetry and determine the order for various Kenyan patterns (Maasai beadwork, tiling designs). A video guides exploration. Lesson 7 — Axis and Order of Rotational Symmetry (3D Extension & Consolidation): Students distinguish between axis of rotational symmetry (for 3D objects) and order of rotational symmetry (for 2D figures). They examine everyday Kenyan objects (a sufuria, a wheel, a gear) and determine their axis and order. Students update cumulative models on the DQB. Lesson 8 — Congruence and Rotation: Students establish formally that rotation is an isometric transformation — it preserves

	length, angle size, and area. They compare rotation to reflection and translation as congruence transformations. A video and worked examples consolidate the link between rotation and congruence. Students argue from evidence: "Are the three turbine blades congruent to each other? Prove it using rotation." Lesson 9 — Applications of Rotation in Real-Life Situations: Students explore and solve problems involving rotation in real-life Kenyan contexts: designing rotationally symmetric logos, calculating the angle swept by a clock hand, the rotation of the Earth, and gear ratios in machinery. Students work in groups to present a real-life rotation problem and solution. Lesson 10 — Unit Consolidation, Model Finalisation & DQB Review: Students finalise their cumulative rotation models, answering the driving question with full evidence. The DQB is reviewed — all initial questions are answered with mathematical justification. Students complete a summative task: given a real-life scenario (turbine blade position), they determine the angle of rotation, centre, order of rotational symmetry, and verify congruence of the image.
Supporting Phenomena	– A video of a Ferris wheel at the Nairobi fun fair rotating about its central axle — students observe individual passenger cabs tracing circular arcs while their distance from the centre remains constant, supporting exploration of how individual points rotate. – Images of Maasai beadwork and Kente-inspired Kenyan fabric patterns showing rotational symmetry — students identify the centre and order of rotational symmetry, connecting the mathematics to local cultural art. – A rotating ceiling fan in a classroom or a video of a gear mechanism from a Kenyan sugar mill (e.g., Mumias Sugar) — supports the lesson on congruence and rotation, showing how rotated images are identical in shape and size to the original. – A time-lapse video or diagram of the Earth rotating about its axis, referenced in the context of Kenyan time zones and sunrise/sunset — supports real-life applications of rotation and the concept of axis of rotation.

LESSON 1 (80 minutes): Anchoring Phenomenon & Introduction to Rotation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To anchor the unit in a real-world phenomenon — the Ngong Hills Wind Farm turbines — and establish the driving question that will guide all lessons in Sub-Strand 3.2: Rotation.
Knowledge	Students will know that rotation is a geometric transformation in which every point on a figure moves through an equal angle about a fixed point called the centre of rotation, and that the shape and size of the figure are preserved.
Skills	Students will be able to identify what stays the same and what changes during a rotation, articulate initial observations about the Ngong Hills turbine phenomenon in written and spoken form, and co-construct a Driving Question Board with meaningful need-to-know questions.
Attitudes	Students will appreciate how mathematics explains real-world phenomena found in Kenya's own landscape, and show curiosity and openness by recording honest predictions before any instruction is given.
Key Inquiry Question	What are the properties of rotation as a geometric transformation?
Purpose in Storyline	This is the opening lesson of the unit. It launches the anchoring phenomenon — the Ngong Hills Wind Farm turbine — and establishes the central puzzle: why do the turbine blades always look the same no matter how far they rotate? Students make initial predictions, gather first observations, and co-construct the Driving Question Board (DQB). Every subsequent lesson will return to this DQB to update understanding and resolve pieces of the puzzle.
Safety Notes	No physical hazards in this lesson. If using a projector or laptop, ensure cables are secured and screens are positioned so all learners can view without straining. Remind students to handle any printed reading materials with care and return them at the end of the lesson.

B. LESSON OVERVIEW
Students are shown a short video of the Ngong Hills Wind Farm turbines rotating. They observe that each blade sweeps through space yet the overall three-blade assembly appears to map perfectly onto itself at every stop. This puzzling observation — movement with no visible change in appearance — launches the unit's driving question. Students write individual predictions, share with partners, and contribute to a whole-class Driving Question Board. They then read a short introductory passage on rotation that introduces key vocabulary (centre of rotation, angle of rotation, direction of rotation, image) without yet providing full explanations, deliberately leaving productive questions open. The lesson closes with students updating their DQB and writing a personal 'What I need to find out' exit card.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Another Pythagorean theorem proof] Source: Khan	Students are told: 'We are going to watch a 90-second clip of the Ngong Hills Wind Farm, located about 25 km from Nairobi in the Rift Valley. Before the video plays, write answers to these two	Display the two prediction questions on the board before the video plays — do NOT show the video yet. Say: 'Do not look anything up. These are your honest first ideas — there are no	Think-Pair-Share (Prediction Version): By requiring written individual predictions before discussion, this strategy externalises prior knowledge and creates a personal stake in the	Circulate and scan exercise books during individual writing. Look for: (a) students who identify 'shape' or 'size' or 'number of blades' as staying the same — these show intuitive grasp of invariance; (b)

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	questions in your exercise book: (1) When a turbine blade moves from one position to another, what do you think stays the same about the whole blade assembly? (2) What do you think changes?' Students work individually in silence for 3 minutes, writing their predictions without consulting peers. After writing, students share predictions with a shoulder partner for 2 minutes using the prompt 'I think _____ stays the same because _____, but I am not sure about _____ .'	wrong answers at this stage.' Circulate silently during individual writing time; do not answer questions. After 3 minutes say: 'Turn to your shoulder partner. You have 2 minutes — use the sentence frame on the board.' After pair-share, cold-call 4 students (select a mix of confidence levels) to share one prediction each. Record key words from each response on the board in a column headed 'Our Predictions'. Say: 'We will come back to check these predictions at the end of the unit.'	phenomenon. Students who encounter evidence that contradicts their prediction are more cognitively engaged in resolving the conflict — this is the foundation of productive struggle for the whole unit.	students who write 'position changes' or 'angle changes' — these show awareness of what rotation does. Flag students who leave both questions blank for targeted support during the Observe phase. Do NOT correct predictions at this stage.
Observe Phase 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher plays a 90-second video clip of the Ngong Hills Wind Farm turbines rotating (sourced from a Kenya Electricity Generating Company / KenGen documentary or similar). Students watch in silence the first time with the instruction: 'Watch the whole clip without writing. Just look.' The clip is played a second time. Students are now given an observation table in their books with three columns: 'What I see moving', 'What appears to stay the same', 'What I am puzzled by'. They complete the table individually during a 5-minute silent observation period after the second viewing. Students then read a one-page introductory passage (provided as a printed handout or displayed page-by-page) titled 'Rotation Around Us — From the Ngong Hills to Your Compass'. The passage introduces: the turbine hub as a fixed point, the concept that each blade traces an arc of equal length, the observation that after rotating by 120° the three-blade assembly looks identical, and real-	Before first viewing say: 'First watch — eyes only, no writing. I want your full attention on the blades.' Play clip. Before second viewing say: 'This time, look specifically at the centre point where the blades meet the hub. Watch one single blade and track it.' Play clip again. After second viewing, give 10 seconds of silence before saying: 'Now open your observation table and fill it in — you have 5 minutes, working alone.' After tables are completed, distribute the reading passage. Say: 'Read the passage once through. Underline any word or idea you have not seen before in a mathematics lesson. Put a question mark next to anything that puzzles you.' Allow 8 minutes for silent reading. After reading, ask: 'Turn to your partner — what is ONE question the passage raised that you cannot yet answer?' Allow 90 seconds. Cold-call 3 pairs to share their questions. Write each question verbatim on the board under the heading 'Questions the Reading Raised'.	Observation Table + Annotated Reading: The two-column structure (what stays the same / what changes) directly trains students to notice the invariance properties of rotation — the core mathematical idea of the unit. The annotated reading strategy (underline new terms, mark questions) activates metacognition: students become aware of gaps in their own understanding, which will motivate engagement with later lessons. Both strategies tie directly to the phenomenon by using the turbine as the referent throughout.	Collect observation tables at the end of the phase (or photograph them). Look for: (a) students who identify the central hub as a fixed point — strong foundation for 'centre of rotation'; (b) students who note that individual blades move but the shape of the three-blade assembly looks the same — strong intuition for rotational symmetry; (c) students who are still describing only colour or size — may need a targeted re-watch with a guiding question. Use cold-call responses to gauge how many students are generating 'why' questions versus only 'what' observations.

	world uses of rotation in Kenyan contexts (clock hands in a Nairobi office, a potter's wheel in Kisumu, a roundabout in Mombasa). The passage intentionally does NOT define 'centre of rotation', 'angle of rotation', or 'direction' formally — it uses everyday language to raise the questions.			
Explain Phase 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students participate in a teacher-facilitated whole-class discussion that builds a shared initial explanation of rotation using the observations and reading from Phase 2. The teacher writes the driving question prominently on the board: 'Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate — and how can we use the mathematics of rotation to predict exactly where any point on the blade will land?' Students contribute ideas to answer: 'What is the fixed point in the turbine? What is changing as the blade moves? Why does the three-blade assembly look the same after the blades move?' The class co-constructs a working definition of rotation in their own words, recorded on the board. The teacher then introduces the formal vocabulary — centre of rotation, angle of rotation, direction of rotation (clockwise / anticlockwise), object and image — and connects each term to the turbine. Students copy the working definition and vocabulary into a dedicated 'Key Ideas' section of their exercise books. The teacher draws a simple diagram on the board: one turbine blade as a line segment from the hub, then shows it rotating to a new position, labelling the centre (hub), the	Open the discussion with: 'From your observation tables and the reading — what is the ONE point on the turbine that never moves?' Wait 10 seconds. Cold-call 2 students. After responses, say: 'In mathematics we call that the centre of rotation. Write that term next to the hub in your diagram.' Repeat the routine for angle: 'What word would you use to describe HOW FAR the blade has moved?' Wait 10 seconds. Cold-call 2 students. Bridge to formal term: 'Mathematicians call this the angle of rotation.' Introduce direction: 'If you are standing in front of the turbine, which way are the blades turning — the same direction as a clock face, or the opposite?' Wait 10 seconds. Cold-call 1 student. Say: 'Clockwise means the same direction as clock hands — negative in our convention. Anticlockwise is positive.' Write convention on board. Draw the board diagram slowly, narrating each label. Say: 'Copy this into your Key Ideas section — you will need it as a reference for every lesson in this unit.' Pause after each label for students to catch up. End with: 'So our working explanation right now is — ' and read the co-constructed definition aloud together.	Concept Mapping via Phenomenon Anchoring: Every new vocabulary term is introduced by pointing back to a visible element of the turbine (hub = centre, blade sweep = angle, direction of spin = clockwise/anticlockwise). This phenomenological anchoring ensures that abstract mathematical vocabulary is never free-floating — it is always attached to a concrete, Kenyan real-world referent that students have already observed. The co-constructed working definition gives students ownership of the explanation.	Ask students to hold up their exercise books showing their labelled diagram. Scan the room for correct labelling of: centre, angle, direction, object (original blade position), image (new blade position). Pose the quick-check question: 'If a blade rotates anticlockwise — is the angle positive or negative?' Cold-call 3 students for individual responses. Students who cannot answer are noted for pair support during Phase 4.

	angle swept, and the direction. Students sketch the diagram in their books.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class co-constructs a Driving Question Board. Each student is given two sticky notes (or small pieces of paper). On the first, they write one question they now have about rotation that they did not have at the start of the lesson. On the second, they write one thing they are now more sure about compared to their original prediction. Students post both notes on the class DQB (a large sheet of paper or section of the classroom wall) divided into three columns: 'What we think we know', 'What we are not sure about', 'What we need to find out'. The teacher facilitates a 5-minute whole-class reading of the DQB, grouping similar questions together and naming the clusters. The teacher maps the clusters to the upcoming lessons: e.g., 'Several of you are asking HOW we measure the angle — we will investigate that in Lesson 2. Others are asking WHY the three-blade assembly maps onto itself — that connects to rotational symmetry, which we will explore in Lesson 4.' Students record in their books: the driving question, and at least two personal 'need-to-know' questions from the DQB.	Distribute sticky notes. Say: 'Think carefully. On your first note write a genuine question — something you are truly puzzled about after today's lesson. On your second note write something you are now more confident about.' Allow 3 minutes of silent individual writing. Say: 'Come up one row at a time and place your notes in the correct column.' Once all notes are posted, spend 5 minutes reading them aloud, grouping similar ones, and annotating clusters with a marker. Say: 'Look at how many of you are asking about the same things — this tells us what we need to investigate together.' Explicitly link each cluster to a future lesson by number. End by reading the full driving question aloud as a class: 'Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate — and how can we use the mathematics of rotation to predict exactly where any point on the blade will land?' Say: 'Every lesson in this unit will bring us closer to fully answering this question. The DQB stays on the wall — we will add to and revise it each lesson.'	Driving Question Board (DQB): The DQB is a living public record of the class's evolving understanding. By physically separating 'what we think we know' from 'what we need to find out', it makes the boundary of current knowledge visible and motivates future lessons as genuine intellectual investigations rather than arbitrary curriculum tasks. Grouping student questions into clusters and mapping them to lessons gives students a sense of agency — their own questions are driving the unit.	Read all sticky notes before the next lesson. Categorise student questions into: (a) questions about vocabulary/definition — addressed in Lesson 2; (b) questions about procedure (how to find centre, how to measure angle) — addressed in Lessons 3–4; (c) questions about the 'why' of rotational symmetry — addressed in Lessons 5–6. Students who post notes in the wrong column or whose notes are very vague ('I don't know anything') are flagged for check-in at the start of Lesson 2.
Model Building Phase 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum)	Students create their first version of a personal Rotation Model in their exercise books. They draw a simple diagram of the Ngong Hills turbine (a circle representing the hub, three line segments representing blades at 120° intervals — though they do not yet know this angle precisely). On the	Say: 'We are going to build a model — a diagram plus an explanation — of what we think rotation is. This model will grow every single lesson. Do not worry about being perfect today; the question marks you put on it are just as important as the labels.' Display the board diagram from	Cumulative Model Building: Starting a personal scientific/mathematical model from Day 1 — even an incomplete one with explicit question marks — commits students to a running record of their understanding. The 'what my model cannot yet explain' prompt is critical: it makes	Collect exit cards. Sort into three piles: (1) Students who correctly use 'centre of rotation' and 'angle' in their Model Statement — ready for Lesson 2 procedural work; (2) Students who use one term correctly but not both — review vocabulary at start of Lesson 2; (3) Students whose Model Statement

🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	diagram they label: the centre of rotation (hub), one blade as 'object', the next blade position as 'image', an arc with an arrow showing direction, and a question mark on the angle label to show it is still unknown. Below the diagram they write a one-sentence 'Model Statement': 'I think rotation moves every point on a shape through the same ________ around a fixed point called the ________.' Students fill in the blanks using today's vocabulary. At the very bottom they write: 'What my model cannot yet explain: ____________.' This model will be revised and enriched in every subsequent lesson.	Phase 3 as a reference. Say: 'Draw your turbine diagram, label everything you can, then write your Model Statement using today's vocabulary words.' Allow 6 minutes. Circulate and prompt: 'What goes in the blank where you wrote the angle? What word did we learn today?' After 6 minutes, cold-call 2 students to read their Model Statement aloud. Say: 'Notice that both models have a question mark on the angle — that is the first mystery we will solve in Lesson 2.' Collect the exit card: 'On a small piece of paper write your name and one thing your model cannot yet explain.' Collect these as students leave.	incompleteness a feature rather than a failure, trains students in the scientific practice of identifying the limits of a model, and generates the intellectual need that drives each subsequent lesson. The turbine diagram connects every revision back to the anchoring phenomenon.	is blank or uses no new vocabulary — meet with these students during the first 5 minutes of Lesson 2 before whole-class instruction begins.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions before planning Lesson 2:

1. PHENOMENON UPTAKE — Did students engage genuinely with the Ngong Hills turbine video, or did the phenomenon feel disconnected from their experience? If engagement was low, consider beginning Lesson 2 with a brief follow-up question: 'Has anyone driven past the Ngong Hills and seen the turbines? What did you notice?'

2. PREDICTION QUALITY — How sophisticated were students' initial predictions? If most predictions were very vague (e.g., 'the blades stay the same'), students may need more structured observation scaffolding in Lesson 2 before moving to formal definitions.

3. DQB RICHNESS — Count the distinct question clusters on the DQB. A rich DQB has at least 4 distinct clusters (e.g., what is a centre of rotation? how do we measure the angle? what is the difference between clockwise and anticlockwise? why do three blades produce this effect?). If fewer than 3 clusters emerged, the reading passage may need to be more provocative — add a worked example with a deliberate mistake for students to catch.

4. VOCABULARY RETENTION — From the exit card scan, what percentage of students correctly used both 'centre of rotation' and 'angle of rotation' in their Model Statement? If fewer than 60% did, open Lesson 2 with a 5-minute vocabulary retrieval activity (e.g., fill-in-the-blank using the turbine diagram) before introducing new content.

5. EQUITY CHECK — Were the same 3–4 students dominating cold-call responses? Review your cold-call list and ensure that in Lesson 2 you deliberately call on students who were silent today, especially those whose observation tables showed strong thinking.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students watched the Ngong Hills Wind Farm turbine video and recorded observations in a three-column table (what is moving, what stays the same, what is puzzling). They noted that individual blades move through space but the overall three-blade assembly appears unchanged at every stop position, and that the central hub point never moves.
What did I learn?	Students were introduced to the vocabulary of rotation: centre of rotation (the fixed point — the turbine hub), angle of rotation (how

	far a point moves around the centre), direction of rotation (clockwise = negative; anticlockwise = positive), object (original position) and image (new position). They co-constructed a working definition: rotation moves every point on a shape through the same angle around a fixed centre, preserving shape and size.
How does this explain the phenomenon?	Students began to explain the anchoring phenomenon: the Ngong Hills turbine blades always look the same because the three-blade assembly is being rotated about its central hub, and rotation preserves the shape and size of the figure. However, students cannot yet explain why exactly 120° produces an identical-looking assembly, how to calculate exactly where any point on a blade will land, or how to determine the centre and angle of rotation from a diagram alone — these questions are recorded on the DQB and will drive Lessons 2–6.

LESSON 2 (80 minutes): Properties of Rotation: What Stays the Same and What Changes?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to investigate and articulate the defining properties of rotation as a geometric transformation using hands-on materials.
Knowledge	By the end of the lesson, the learner should be able to: (a) identify the centre of rotation as the fixed point about which a shape turns; (b) state that distances from the centre of rotation are preserved under rotation; (c) state that the object and image are always congruent under rotation; (d) distinguish between positive (anti-clockwise) and negative (clockwise) rotation angles; (e) recognise that an angle of rotation is measured at the centre from a point on the object to the corresponding point on the image.
Skills	Measuring angles using a protractor; rotating cut-out shapes manually about a fixed centre; recording and comparing distances from centre to object and image points; communicating mathematical observations precisely.
Attitudes	Curiosity and patience in carrying out repeated measurements; appreciation that mathematical properties discovered in the classroom explain real-world phenomena such as the Ngong Hills wind turbine blades.
Key Inquiry Question	What are the properties of rotation as a geometric transformation?
Purpose in Storyline	In Lesson 1 students established that the Ngong turbine blades appear unchanged after every rotation. In this lesson students gather hands-on evidence that explains WHY this is so: rotation preserves distances from the centre, preserves shape and size (congruence), and is fully described by a centre, an angle, and a direction. This resolves the first layer of the driving question — 'why do the blades always look the same?' — and sets up Lesson 3, where students will locate the exact centre and measure the angle of rotation algebraically.
Safety Notes	Scissors are used to cut out shapes — remind learners to handle scissors with care, cutting away from the body. Ensure protractors and rulers are not used as projectiles. Learners with visual impairments should be paired with a sighted partner and given tactile/raised-line protractors where available.

B. LESSON OVERVIEW

Learners begin by revisiting the Ngong Hills wind turbine phenomenon and the Driving Question Board from Lesson 1. In small groups they rotate cut-out cardboard blade shapes about a pinned centre on grid paper, use rulers to measure distances from centre to corresponding points on object and image, and use protractors to measure the angle swept. They discover that: (1) the centre pin never moves; (2) every point on the object is the same distance from the centre as its image; (3) the object and image are congruent; and (4) turning anti-clockwise gives a positive angle while turning clockwise gives a negative angle. Groups record findings on a Properties-of-Rotation evidence table and share with the class. A whole-class sensemaking discussion connects every finding back to the turbine blades. Learners update their Driving Question Board and revise their cumulative rotation model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown a still image of one Ngong Hills turbine blade	Display the frozen turbine image on the ARES screen. Say: 'In	Think-Pair-Share Prediction: activates prior knowledge about	Observable indicator: Every learner has written at least one

📹 **VIDEO:** [Another Pythagorean theorem proof] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	assembly frozen mid-rotation. Each learner silently writes answers to three prompt questions in their exercise books: (1) 'If I pin a cut-out triangle at one corner and turn it, which parts of the triangle do you predict will change and which will stay the same?' (2) 'Do you predict the distance from the pin to any corner of the triangle will change after turning? Why?' (3) 'If I turn the shape to the left (anti-clockwise) vs to the right (clockwise), do you think the result will be mathematically different?' Learners then share predictions with a shoulder partner using a Think-Pair-Share routine before two cold-called pairs share with the class. Predictions are recorded on a class 'Prediction Anchor Chart' that will be revisited at the end of the lesson.	Lesson 1 we saw that the Ngong turbine blades look identical no matter where they stop. Today we are going to find out exactly WHY. But first — predict.' Distribute the prediction prompt slip (or read prompts aloud). Allow 3 minutes of silent individual writing. Then say: 'Turn and tell your shoulder partner your prediction — you have 90 seconds.' After paired discussion, say: 'I am going to cold-call two pairs to share.' [WAIT 10 seconds after posing each question before cold-calling.] Cold-call Pair 1: 'What did you predict would stay the same?' Cold-call Pair 2: 'What did you predict about distance from the pin to a corner?' Record key prediction words (e.g., 'size stays same', 'distance changes', 'direction matters') verbatim on the Prediction Anchor Chart. Say: 'Hold onto these predictions — we will come back and check them with evidence.'	shape transformations, creates cognitive dissonance when predictions conflict with evidence, and gives the teacher immediate insight into misconceptions (e.g., learners who believe rotation changes size or distance from centre).	prediction per prompt in their book before paired sharing begins. Teacher circulates and scans books during the 3-minute silent write — look for the misconception 'the distance from centre to image point is different from the distance to the object point.' Flag 2–3 learners holding this misconception to cold-call during Phase 3 to address it directly.
Observe Phase 📹 **VIDEO:** [Another Pythagorean theorem proof] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	In groups of three, learners receive: one cardboard cut-out of a three-bladed turbine shape (or equilateral triangle labelled A, B, C), one sheet of A3 grid paper, a drawing pin or sharp pencil as a pivot, a protractor, a ruler, and a pencil. Step 1 — Learners place the cut-out on the grid paper and trace around it (this is the object). They mark the centre pin position as point O. Step 2 — Learners measure and record the distance OA, OB, OC (from centre to each vertex of the object) in a results table. Step 3 — Learners rotate the cut-out 90° anti-clockwise about O (using the protractor to confirm the angle). They trace the new position (this is Image 1) and label the image vertices A', B', C'.	Distribute materials. Say: 'Your job is to be like a Kenyan engineer at Ngong Hills — measure carefully and record everything. Your evidence table is your data log.' Demonstrate Steps 1–3 at the front using a large cut-out pinned to the board, narrating: 'I am placing my protractor at O, lining up the zero line with OA, and turning anti-clockwise until I reach 90 degrees — that is a positive rotation.' Circulate continuously. At each group, ask: 'What do you notice about OA and OA'?' [WAIT 12 seconds for a response before prompting.] If a group finds OA ≠ OA', say: 'Check your ruler placement — is it starting exactly at O?' After 25 minutes of group work, call time. Cold-call 3 different	Structured Data Collection with Evidence Table: learners gather quantitative evidence (measured distances and angles) rather than relying on visual impression alone. Recording in a table makes the invariant property (OA = OA') visible as a pattern across multiple rotations and multiple groups, building the empirical case for the formal property.	Observable indicator: Each group's evidence table has at least two completed rows (two rotations) with measured values for OA, OB, OC and OA', OB', OC'. The sentence describing the pattern uses the words 'same' or 'equal' in relation to distance. Teacher checks for groups who have measured from an incorrect origin point O — correct immediately by modelling at their table.

	Step 4 — Learners measure OA', OB', OC' and record in the same table. Step 5 — Learners rotate again by −120° (clockwise) to produce Image 2 and repeat measurements. Step 6 — Learners place the object and Image 1 on top of each other and check whether they match perfectly (congruence check). Step 7 — Learners complete the evidence table: 'Distance from O to object vertex' vs 'Distance from O to image vertex' for each rotation, and write one sentence describing what they notice.	groups (one per rotation attempted) to read their OA and OA' values aloud. Write values on the board in a class results table. Ask: 'What pattern do you see across all our groups?' [WAIT 15 seconds.] Cold-call 2 learners who have not yet spoken.		
Explain Phase 📹 **VIDEO:** [Another Pythagorean theorem proof] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Whole-class discussion facilitated by the teacher draws out four formal properties from the evidence. Learners then individually write the four properties in their own words in their exercise books, alongside a labelled diagram showing: centre O, object point A, image point A', angle of rotation, and the equal distances OA = OA'. Next, learners apply the properties to a short guided practice: given a grid with a kite shape (labelled with Kenyan context — 'a kite flown at Uhuru Park, Nairobi') and a marked centre O, learners (a) predict where vertex P will land after a +90° rotation, (b) confirm using a protractor and ruler, and (c) write whether the kite and its image are congruent and why. Learners reconnect findings to the turbine: 'Use the four properties to explain in two sentences why each Ngong turbine blade looks the same after every rotation.'	Open the class discussion by pointing to the class results table from Phase 2. Say: 'Our data shows something powerful. Who can tell me what is true about OA and OA' in every single row?' Cold-call 2 learners. After responses, say: 'Yes — the distance from the centre to any point on the object is exactly equal to the distance from the centre to the corresponding point on the image. This is Property 1: rotation preserves distance from the centre.' Write on board: 'Property 1: OA = OA' (distance from centre is invariant).' Proceed through all four properties, eliciting each from learners before formalising: Property 2 — object and image are congruent (same shape and size); Property 3 — the centre of rotation is fixed (does not move); Property 4 — the direction of rotation matters: anti-clockwise = positive angle, clockwise = negative angle. For Property 4, physically demonstrate with the large cut-out: 'I turn this way [anti-clockwise] — that is +90°. I turn back [clockwise] — that is −90°. These are different	Evidence-to-Property Bridging Discussion: the teacher uses the class data table as the anchor for formalising each property, ensuring that abstract mathematical language (invariant, congruent, centre, angle, direction) is directly tied to measurements learners themselves collected. This prevents rote memorisation and builds transferable conceptual understanding.	Observable indicator: Learner books contain four labelled properties AND a correctly labelled diagram with O, A, A', angle arc, and the equality OA = OA' marked. For the kite task, the correct image vertex position is within 2° and 2 mm of the mathematically accurate position. The turbine explanation sentences explicitly name at least two properties (e.g., 'congruent', 'distance preserved', 'fixed centre', 'direction').

		rotations.' For the Uhuru Park kite task, circulate for 8 minutes. [WAIT 12 seconds after posing each guided practice question.] Cold-call 3 learners to share their turbine explanation sentences. Ask the class: 'Do you agree? Does this sentence use at least two of the four properties?' [WAIT 10 seconds.] Invite a peer to add or improve the explanation.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed on the ARES screen (established in Lesson 1). Each learner receives two sticky-note-sized paper slips. On Slip 1 (green — 'Evidence Gathered'), learners write one piece of evidence from today's lesson that helps answer the driving question: 'Why do the blades always look the same no matter how far they rotate?' On Slip 2 (orange — 'New Question'), learners write one new question today's lesson has raised for them (e.g., 'How do we find the centre if it is not already marked?' or 'What if the angle is more than 360°?'). Learners post their slips on the DQB under the correct columns. The class does a 3-minute gallery walk of the DQB, then two learners are cold-called to read aloud a green slip and two to read an orange slip. The teacher clusters orange slips into themes and labels them as targets for upcoming lessons.	Say: 'Let us go back to our big question about the Ngong turbine. Look at the Driving Question Board from Lesson 1. Think about what new evidence you have today.' Distribute the two slips. Allow 4 minutes of silent writing. Say: 'Post your slips — green on the left, orange on the right.' Facilitate the 3-minute gallery walk: 'Walk silently, read your classmates' evidence and questions, and give a tick to any slip you agree with.' After the walk, cold-call 2 learners: 'Read one green slip you ticked and explain why you agree with it.' Then cold-call 2 different learners: 'Read one orange slip — do you think you already know the answer, or is it a genuine mystery?' Cluster orange slips by theme (e.g., 'finding the centre', 'angles greater than 360°', 'negative angles'). Say: 'These orange questions are exactly what the next lessons will answer — keep them in mind.'	Driving Question Board Update with Gallery Walk: making evidence and new questions publicly visible allows learners to see the collective growth of the class's understanding, validates individual contributions, and creates a living 'knowledge map' that motivates future lessons by showing what remains to be resolved.	Observable indicator: Every learner has posted both a green (evidence) slip and an orange (new question) slip. Green slips must reference at least one named property (e.g., 'congruent', 'fixed centre', 'distance preserved') rather than a vague statement like 'shapes are the same'. Teacher collects all slips at the end of the lesson to check for quality and to inform Lesson 3 planning.
Model Building Phase 📹 **VIDEO:** Another Pythagorean theorem proof	Learners return to the cumulative rotation model begun in Lesson 1 (a labelled sketch of the Ngong turbine with annotations). In their exercise books they revise and extend the model by: (a) labelling the hub as 'Centre of Rotation O	Say: 'Open your model from Lesson 1. Today's evidence means we need to make it more accurate and more mathematical. Add five specific annotations — I will read them out one at a time and give you 2 minutes for each.'	Cumulative Annotated Model Revision: iteratively building and revising a single model across all 10 lessons makes abstract properties tangible and shows learners how mathematical understanding accumulates. Peer	Observable indicator: The revised model contains all five required annotations, each using correct mathematical vocabulary. The equal-length radii are drawn from O (not from a blade edge). The curved angle arrow originates at

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	— fixed, does not move'; (b) drawing arrows from O to two blade tips and labelling both arrows with equal lengths and the notation 'OA = OA' (distance preserved)'; (c) adding a curved arrow showing the angle of rotation and labelling it with a positive sign for anti-clockwise; (d) writing the word 'CONGRUENT' between the original blade position and the rotated blade position with a brief justification ('same shape, same size — only position changed'); (e) annotating: 'Positive angle = anti-clockwise turn; Negative angle = clockwise turn.' Groups share their revised models with an adjacent group and give one 'strength' and one 'add-on suggestion' as peer feedback. Learners incorporate peer feedback before the lesson closes.	Read each annotation task slowly, modelling on the board's large turbine diagram. Circulate and check that: the label 'Centre O — fixed' is placed at the hub; two equal-length radii are drawn and labelled; the curved angle arrow is correctly placed at O (not at the blade tip); 'CONGRUENT' appears with a justification sentence; and direction notation (positive/negative) is present. After individual revision, say: 'Share your model with the group next to you. Give them one thing they did well and one mathematical detail they could add. Use mathematical vocabulary: centre, angle, congruent, distance, direction.' [WAIT 10 seconds for groups to begin.] After peer feedback (5 minutes), say: 'Make your final addition now.' Close by saying: 'Your model is now explaining the science behind a real Kenyan wind farm. Every time an engineer at Ngong Hills checks a turbine, these four properties are at work.'	feedback during model sharing reinforces precise mathematical vocabulary and reveals gaps in understanding before the next lesson.	O. Peer feedback slips contain at least one subject-specific mathematical term. Teacher photographs 4–5 models (with learner permission) to display on the ARES platform as exemplars for Lesson 3.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did every group successfully measure OA = OA' within an acceptable margin of error (≤ 2 mm)? If not, was the issue with protractor alignment, ruler placement, or understanding of where O is? Plan a brief correction demo at the start of Lesson 3. (2) Which property caused the most confusion — the distinction between positive (anti-clockwise) and negative (clockwise) angles is historically the most common misconception at this level. Count how many orange DQB slips referenced direction or sign of angle — if more than one-third of the class raised this, open Lesson 3 with a dedicated 5-minute direction drill. (3) Were learners able to connect the four properties back to the Ngong turbine in their own words, or did their turbine explanation sentences remain vague? Review the sentences written in Phase 3 tonight and identify 2–3 strong exemplars to share anonymously at the start of Lesson 3 as a model. (4) Did the peer feedback during model sharing use mathematical vocabulary (congruent, invariant, centre, angle, direction), or was it purely descriptive ('it looks nice')? If vocabulary use was weak, introduce a 'sentence starter card' for peer feedback in Lesson 3: 'One mathematical strength is… because… One thing to add is…'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners rotated cardboard cut-out shapes (modelled on the Ngong Hills turbine blades) about a fixed pin on grid paper. They measured the distances from the centre O to each vertex of the object and to the corresponding vertex of the image for rotations of

	+90° (anti-clockwise) and −120° (clockwise), and recorded all values in an evidence table. They also performed a congruence check by overlaying the object and image cut-outs.
What did I learn?	Rotation has four defining properties: (1) the centre of rotation O is fixed and does not move; (2) the distance from O to any point on the object equals the distance from O to its image — OA = OA' (distance is invariant); (3) the object and its image are always congruent — same shape, same size, only position changes; (4) direction matters — a positive angle means anti-clockwise rotation and a negative angle means clockwise rotation. The angle of rotation is measured at the centre, from the line joining O to the object point to the line joining O to the image point.
How does this explain the phenomenon?	These four properties explain why the Ngong Hills wind turbine blades always look the same no matter where they stop: the hub is the fixed centre of rotation, each blade tip is the same distance from the hub before and after turning (distance invariant), and the entire three-blade assembly is congruent to itself after every rotation — the blades have simply changed position, not shape or size. The direction of spin (positive = anti-clockwise when viewed from the front) is a mathematical fact about the rotation that engineers can measure and predict precisely.

LESSON 3 (80 minutes): Rotating Points on the Cartesian Plane	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to rotate individual points by 90°, 180°, and 270° (clockwise and anticlockwise) about the origin on the Cartesian plane using coordinate rules, and to verify their results using a simulation.
Knowledge	By the end of the lesson, the learner should be able to: (a) state and apply the coordinate rules for rotating a point by 90°, 180°, and 270° about the origin; (b) distinguish between clockwise (negative) and anticlockwise (positive) rotation directions; (c) identify the origin as the centre of rotation and describe what remains fixed during rotation.
Skills	By the end of the lesson, the learner should be able to: (a) accurately plot the image of a point after a given rotation on the Cartesian plane; (b) derive coordinate transformation rules by observing patterns across multiple rotations; (c) verify rotation images using a digital simulation and reconcile any discrepancies.
Attitudes	The learner appreciates that precise mathematical rules — not guesswork — govern how points move under rotation, connecting this precision to how the Ngong Hills wind turbine blades sweep through predictable, repeatable positions.
Key Inquiry Question	How do we determine the centre and angle of rotation given an object and its image?
Purpose in Storyline	In Lessons 1 and 2 students established what rotation is, identified the centre of rotation, and explored angle and direction. Lesson 3 now equips students with the algebraic coordinate rules that let them PREDICT — with exact numbers — where any point lands after a 90°, 180°, or 270° rotation about the origin. This directly advances the driving question: if we can write a rule for any point on a turbine blade, we can predict every position the blade occupies as it spins.
Safety Notes	No physical hazards. Ensure screen brightness is comfortable when using the simulation. Learners sharing devices should handle them responsibly and return them to the charging station at the end of the lesson.

B. LESSON OVERVIEW
Learners begin by predicting what happens to the coordinates of a point on a Ngong Hills turbine blade when the blade rotates 90°, 180°, and 270° about the central hub (modelled as the origin). They then systematically rotate several points on the Cartesian plane by hand, record the original and image coordinates in a table, and look for a pattern. A short video simulation confirms or challenges their pattern. Through structured sensemaking, learners derive and articulate the four standard rotation rules (90° anticlockwise, 90° clockwise, 180°, 270° anticlockwise). They apply the rules to a multi-point problem set in the context of a turbine blade tip, then update the class Driving Question Board with coordinate evidence about what changes and what stays the same during rotation.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Intro to the coordinate plane]	Learners are shown a still image of the Ngong Hills wind turbine with one blade tip labelled as point P(4, 2) on a Cartesian grid overlaid on	Display the turbine image and say: 'You are an engineer at KenGen monitoring the Ngong Hills wind farm. The blade tip is at P(4, 2). As	Think-Pair-Share with Prediction Table — students externalise prior knowledge and surface misconceptions (e.g., confusing	Observable indicator: Each learner has a completed three-column prediction table with at least one coordinate pair written for each of

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the hub at the origin. Working individually in their exercise books, each learner writes down their prediction: 'If the blade rotates 90° anticlockwise about the hub, where do you think point P will land? Write the coordinates.' After 3 minutes, learners share predictions with a neighbour (Think-Pair). Selected pairs share with the class. A second prompt follows: 'Now predict where P lands after 180°, and after 270° anticlockwise.' Learners record all three predictions in a three-column table (Rotation | My Predicted Image | Actual Image — the last column left blank). The teacher records a sample of class predictions on the board without affirming or correcting them, creating productive tension.	the blade spins anticlockwise, you need to predict exactly where it will be after a quarter turn, a half turn, and a three-quarter turn. Write your prediction NOW — no discussion yet.' Allow 3 minutes of individual silent work. After 3 minutes say: 'Turn to your neighbour — compare your predictions. Do you agree? If not, who do you think is closer and why? You have 2 minutes.' Cold-call 3 pairs to share: 'Pair at the front — what did you predict for 90°?' Record responses on the board verbatim (e.g., 'Some say (−2, 4), some say (2, −4), some say (4, 2)'). Say: 'We have disagreement — that is exactly where mathematics becomes useful. By the end of this lesson you will know with certainty.' Do NOT resolve the disagreement yet. WAIT TIME: After each cold-call question, wait 10 seconds before accepting a response.	clockwise and anticlockwise, assuming coordinates do not change). The unresolved class disagreement recorded on the board creates cognitive dissonance that motivates the Observe phase.	the three rotations. Teacher circulates and scans tables — note which misconceptions are most common (e.g., sign errors, x/y swap errors) to target during the Explain phase. Red flag: any learner who writes the same coordinates for all three rotations (indicating no understanding of transformation).
Observe Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Hands-on coordinate investigation (12 min): Learners work in pairs. Each pair receives a printed Cartesian grid (or draws one in their book, axes from −6 to 6). They plot the point A(3, 1) and physically rotate it step by step: (i) 90° anticlockwise about the origin — they use a ruler and protractor or the 'quarter-turn' method (rotate the page 90°) to locate the image A', then read off coordinates; (ii) 180° — locate A''; (iii) 270° anticlockwise — locate A'''. They record results in a table: Original | 90° ACW | 180° | 270° ACW. They repeat for a second point B(−2, 3). They look for a pattern and write a conjecture: 'I think the rule for 90° anticlockwise is…'. Part B — Simulation verification (8 min): The	Distribute grids. Say: 'You are going to find the pattern mathematically — like engineers calibrating the turbine sensor positions. Plot A(3,1). Now rotate your whole page 90° anticlockwise — where does A land? Read the coordinates and record them. Do the same for 180° and 270°.' Circulate actively. After 4 minutes say: 'Now try point B(−2, 3). Same process.' After another 4 minutes, cold-call 2 pairs: 'What coordinates did you get for A after 90° anticlockwise?' Write responses on the board. Then say: 'Now look at your table — what do you NOTICE about what happened to the x and y values?' Wait 10 seconds. Cold-call 3 learners. Before playing the simulation say: 'Watch carefully —	Data Table Pattern Recognition — learners generate their own data (coordinates from physical rotation) and look for structure in the numbers before being told the rule. This mirrors how Kenyan mathematician-engineers derive formulae from observed data rather than memorising rules in isolation. The simulation serves as an authoritative secondary source that either confirms or constructively challenges their hand-drawn results.	Observable indicator: Learner tables contain correct image coordinates for at least 3 of the 4 rotation cases for both points A and B. Conjectures written in their own words (even if imprecise) show an attempt to describe x/y swapping and sign changes. Teacher listens for pairs who articulate 'the x becomes the negative y' — invite them to share in Phase 3. Red flag: pairs who have incorrect values for both points — pull them aside during Phase 3 for targeted re-teaching.

	teacher plays a 4-minute ARES offline simulation/video showing a point rotating about the origin in 90° steps, with coordinates displayed at each stop. Learners check their conjecture against the simulation and fill in the 'Actual Image' column of their prediction table from Phase 1. They also note: 'Positive rotation = anticlockwise; negative rotation = clockwise' as a key observation.	every time the point jumps, read the new coordinates aloud in your head. Then check your prediction table.' After the simulation, say: 'Update your Actual Image column now. Then write — did the simulation agree with your hand calculation? If not, where did your error occur?'		
Explain Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners consolidate their observations into formal coordinate rules through a structured class discussion. The teacher guides them to articulate and record the four rules in their books: (1) 90° anticlockwise: (x, y) → (−y, x); (2) 90° clockwise / 270° anticlockwise: (x, y) → (y, −x); (3) 180° (either direction): (x, y) → (−x, −y); (4) 270° anticlockwise / 90° clockwise confirmed as equivalent. Learners connect each rule back to the turbine: 'After a 90° anticlockwise turn of the blade, the tip that was at (4, 2) is now at (−2, 4).' They then work through three guided practice problems individually: P1 — Rotate point T(5, −3) by 90° anticlockwise; P2 — Rotate point K(−4, 1) by 180°; P3 — Rotate point M(2, 6) by 270° anticlockwise. After each problem, learners compare with a neighbour before the teacher reveals the answer. The teacher also reinforces: 'The Earth rotating on its own axis is another example — as it spins, a city like Nairobi moves through space, but the axis stays fixed, just like our origin.'	Begin by writing the class data from the board (from Phase 2) in a neat table. Say: 'Let's read the pattern together. For 90° anticlockwise, A(3,1) became — what?' Cold-call a learner. Write (−1, 3). Say: 'And B(−2, 3) became?' Cold-call another learner. Write (−3, −2). Say: 'Look at the original x and y. Look at the image. What operation connects them? Talk to your neighbour for 30 seconds.' After 30 seconds, cold-call 3 pairs. Guide them to articulate: 'The new x is the negative of the old y; the new y is the old x.' Write the rule formally: '90° ACW: (x, y) → (−y, x).' Repeat the process for 180° and 270°. For 270° say: 'A 270° anticlockwise turn — how many degrees clockwise is that? Think for 10 seconds.' WAIT 10 seconds. Cold-call: 'Yes — 90° clockwise. So these two rules are the same.' For practice problems, say: 'Problem 1 — T(5, −3), 90° anticlockwise. Apply the rule — WRITE the answer, do not call out.' Allow 90 seconds. Then: 'Show your answer to your neighbour. Do you agree?' Cold-call one pair to justify step by step. Say: 'Always show the substitution: new x = −(−3) = 3, new y = 5, so image is (3, 5).'	Rule Derivation from Data + Worked Example Justification — learners arrive at the coordinate rules through their own pattern recognition rather than being told. This develops mathematical reasoning. Connecting to the Earth's rotation and Nairobi's daily journey through space contextualises abstract algebra within a familiar Kenyan phenomenon, reinforcing that these are not arbitrary rules but descriptions of real physical motion.	Observable indicator: Learners can correctly apply the stated rule to a new point without referring to a grid (they use algebra alone). For P1: correct answer is (3, 5); P2: (4, −1); P3: (−6, 2). Teacher marks these by circulating during the 90-second work window. Any learner with 2 or more wrong answers receives a prompt card with a worked example to reference. Red flag: systematic sign errors (e.g., always getting the x sign wrong) indicate a misremembering of which coordinate changes sign — address with the mnemonic 'Swap then flip the first.'

		Repeat for P2 and P3.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board. In pairs, they write one sticky note (or a card) responding to each of the following prompts: (a) EVIDENCE CARD — 'One coordinate rule I can now use to predict where a turbine blade point lands is…'; (b) STILL WONDERING — 'Something I am still unsure about is…'; (c) CONNECTION TO THE PHENOMENON — 'This connects to the Ngong Hills turbine because…'. Cards are placed on the DQB under the appropriate columns. The teacher reads out 3–4 cards aloud and highlights how today's coordinate evidence advances the driving question: 'Now we don't just say the blade looks the same — we can PROVE it with numbers.' The teacher also adds a new question to the DQB for the class: 'What happens when we rotate a whole shape — not just one point — about the origin? Does every point follow the same rule?'	Say: 'In Lessons 1 and 2 we established that rotation has a centre, a direction, and an angle. Today we added something new — coordinate RULES. Let's update our board.' Distribute sticky notes or index cards. Say: 'Write your evidence card first — one sentence, one rule, with a specific example using turbine coordinates. You have 2 minutes — go.' After 2 minutes: 'Now your Still Wondering card — be honest, this is not marked, it helps us plan the next lesson.' After 2 more minutes, have pairs walk to the board and place their cards. Read 3 Evidence cards aloud. Say: 'Listen — [learner name]'s group wrote that after a 90° ACW rotation the tip at (4,2) moves to (−2, 4). Does everyone agree? Thumbs up if yes.' Then point to the new teacher question on the DQB and say: 'This is what we will investigate in Lesson 4. Write it in your books as our next challenge.'	Driving Question Board — Three-Column Card Sort (Evidence / Still Wondering / Phenomenon Connection). This metacognitive routine makes learners' growing understanding visible and communal. It also surfaces residual confusion that informs lesson planning, and maintains the narrative thread back to the Ngong Hills phenomenon so that coordinate algebra never feels decontextualised.	Observable indicator: Every pair produces at least one accurate Evidence card that states a correct coordinate rule and applies it to a specific turbine-context coordinate (not a generic example). 'Still Wondering' cards that mention shapes, multiple points, or direction confusion indicate readiness targets for Lesson 4. Red flag: Evidence cards that state vague rules like 'the numbers change' without specifying how — these pairs need targeted follow-up at the start of Lesson 4.
Model Building Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook)	Learners revise and extend their individual rotation model. Each learner has a 'Rotation Model Sheet' started in Lesson 1 (a diagram of the turbine with annotations). Today they add: (1) A Cartesian grid overlaid on the turbine hub at the origin, with one blade tip labelled P(4, 2) and its images after 90°, 180°, and 270° anticlockwise rotation plotted and labelled P', P'', P''' with coordinates written next to each; (2) A 'Rules Box' listing all four coordinate transformation rules with arrows showing direction; (3) An updated explanatory sentence: 'The turbine	Say: 'Open your Rotation Model Sheet. Today you are adding coordinate precision to your model. This is what separates an engineer's technical drawing from a rough sketch — exact coordinates.' Allow 10 minutes for model updating. Circulate and ask individual learners: 'Show me where P(4,2) is on your grid. Now show me P' after 90° ACW — what rule did you use? What are the coordinates?' Provide corrective feedback immediately: 'Check your rule — new x = negative of old y. What is the negative of 2? So new x is −2, correct.' After 10 minutes,	Cumulative Model Revision — learners do not create a new diagram each lesson but iteratively annotate and extend the SAME model, mirroring how scientific and engineering models are refined as evidence accumulates. The addition of the Rules Box alongside the visual grid integrates geometric and algebraic representations, building dual coding of the concept. The extension question (360° rotation) plants a seed for rotational symmetry without over-teaching.	Observable indicator: The updated model contains a correctly labelled Cartesian grid with P(4,2) and its three images plotted at correct coordinates: P'(−2,4), P''(−4,−2), P'''(2,−4). The Rules Box lists at minimum the 90° ACW rule with correct algebraic notation. The explanatory sentence references both the rule and the return-to-start property. Red flag: Models where images are plotted in the wrong quadrant indicate a persistent confusion between clockwise and anticlockwise — record names for re-teaching at lesson start of Lesson 4.

Source: CK-12 🔍 Search ARES for similar readings	blade looks the same after each 90° rotation because each point maps to a new position following the rule (x,y)→(−y,x), and after four such rotations every point returns exactly to where it started.' Learners who finish early are challenged to mark: 'What is the image of the tip after a 360° anticlockwise rotation — and why?' to deepen their understanding of rotational symmetry as a preview of Lesson 4.	cold-call one learner to project or display their model and explain it to the class in 2 minutes. Close by saying: 'Your model now has geometry AND algebra. In Lesson 4 we will rotate entire shapes — all the points of the blade at once — and you will use exactly these rules to do it. Keep your model safe.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) EVIDENCE OF RULE ACQUISITION — Could learners state the 90° anticlockwise rule (−y, x) without looking at their notes by the end of Phase 3? If fewer than 70% could do so, plan a 5-minute retrieval practice opener for Lesson 4 before introducing shape rotation. (2) DIRECTION CONFUSION — Did learners consistently mix up clockwise and anticlockwise? If yes, introduce a physical kinesthetic anchor at the start of Lesson 4: stand up and physically rotate your body 90° anticlockwise — note which way your face turns. (3) DQB QUALITY — Were the Evidence cards specific (contained actual coordinate pairs) or vague? Vague cards signal that learners understand the rule procedurally but cannot yet connect it to the phenomenon — use a turbine context warm-up in Lesson 4. (4) MODEL COMPLETENESS — Were the three image points plotted in correct quadrants? Quadrant errors (especially P'' in the wrong quadrant) suggest sign-rule confusion for 180° — re-examine using the Earth/Nairobi rotation analogy: after half a spin, Nairobi is on the opposite side of the axis, so both coordinates flip sign. (5) EXTENSION ENGAGEMENT — How many learners attempted the 360° question? High engagement suggests the class is ready to explore rotational symmetry earlier than planned; low engagement suggests consolidation is the priority for Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students watched a simulation and used Cartesian grids to physically rotate points A(3,1) and B(−2,3) through 90°, 180°, and 270° anticlockwise about the origin, recording image coordinates in a data table and comparing hand-plotted results with the digital simulation output.
What did I learn?	Students derived and applied the four coordinate rotation rules about the origin: 90° anticlockwise (x,y)→(−y,x); 90° clockwise / 270° anticlockwise (x,y)→(y,−x); 180° (x,y)→(−x,−y); and confirmed that a positive rotation angle means anticlockwise while a negative angle means clockwise.
How does this explain the phenomenon?	Students explained that the Ngong Hills turbine blade tip at P(4,2) moves to predictable, calculable positions — (−2,4), (−4,−2), and (2,−4) — after successive 90° anticlockwise rotations, and that after four such rotations it returns exactly to (4,2), beginning to explain mathematically why the blade always looks the same regardless of its rotational position.

LESSON 4 (80 minutes): Rotating Shapes on the Cartesian Plane	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To extend rotation from single points to full shapes (triangles and quadrilaterals) on the Cartesian plane, rotating about the origin and other given centres, and to verify properties of the image.
Knowledge	Students will know that: (1) every vertex of a shape rotates through the same angle about the same centre; (2) rotating all vertices of a shape and joining them in order produces the image; (3) the image is congruent to the object and the same distance from the centre of rotation; (4) positive rotation angles indicate anticlockwise turns and negative angles indicate clockwise turns.
Skills	Students will be able to: (a) rotate a given triangle or quadrilateral about the origin through 90°, 180°, and 270° (both anticlockwise and clockwise) using a protractor, ruler, and squared paper; (b) rotate a shape about a centre of rotation other than the origin; (c) read off and record the coordinates of image vertices; (d) verify congruence of object and image by measuring side lengths and angles; (e) use a digital tool (GeoGebra) to check constructions.
Attitudes	Students will appreciate the precision required in geometric construction and value the use of both manual and digital tools as complementary methods for verifying mathematical results.
Key Inquiry Question	How do we rotate a complete shape on the Cartesian plane, and what properties of the shape are preserved after rotation?
Purpose in Storyline	In Lessons 1–3 students established what rotation is, identified its key elements (centre, angle, direction), and practised rotating single points. In Lesson 4 students scale up to rotating full shapes — just as every point on a turbine blade rotates through the same angle about the same hub. This lesson directly answers the sub-question: 'If every point on a blade moves, why does the overall blade shape look exactly the same?' By rotating triangles and quadrilaterals and measuring their images, students gather the evidence that rotation preserves shape and size — a key piece of the Ngong Hills puzzle.
Safety Notes	Remind students to handle geometrical instruments (compasses, protractors with sharp edges) carefully. Students should not share pencils during individual construction tasks to maintain focus. Ensure adequate desk space so that large sheets of squared paper do not cause collisions.

B. LESSON OVERVIEW

Students begin by predicting what will happen to a triangle drawn on the Cartesian plane when it is rotated 90° anticlockwise about the origin, sketching their guesses before any instruction. They then carry out a guided hands-on construction activity — rotating a triangle and a quadrilateral about the origin and about a given non-origin centre — using protractors, rulers, and squared paper. Each group records the object and image coordinates in a table and measures side lengths and angles to test congruence. A short GeoGebra verification session allows students to check their paper constructions digitally. In the Explain phase, the class co-constructs a rule for how coordinates transform under 90°, 180°, and 270° rotations about the origin. Students then update the Driving Question Board with new evidence and revise their cumulative model of the turbine blade system, annotating it with congruence marks and rotation arcs to show that every point on the blade traces an equal arc about the fixed hub.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student receives a printed Cartesian grid showing triangle KMN with K(1,2), M(3,2), N(3,5) — labelled 'Blade Tip Triangle' to keep the turbine context alive. Working individually in silence, students sketch where they think each vertex will land after a 90° anticlockwise rotation about the origin. They label their predicted image K′M′N′ and write one sentence below: 'I think this because…'. They also answer: 'Do you think the rotated triangle will be the same size and shape as the original? Explain.' Predictions are not discussed yet — students fold the paper and place it face-down to return to after the Observe phase.	Distribute grids and say: 'On your grid you can see triangle KMN — think of it as a cross-section of one blade on the Ngong Hills turbine. Without talking to your neighbour, sketch where you think each vertex will land after a 90° anticlockwise rotation about the origin — the turbine hub.' Allow 4 minutes of individual silent work. Circulate and note 3–4 different prediction types on a sticky note for later comparison (e.g. reflections across axes, correct rotations, rotations of wrong magnitude). Do NOT correct errors yet. After 4 minutes, cold-call 3 students: 'Amina, tell me one vertex — where did you put K′?' Record the three predictions as quick sketches on the board under the heading 'Our Predictions'. Say: 'Hold on to your reasons — we will return to these very soon.'	Prediction with Justification — by committing a prediction to paper before instruction, students activate prior knowledge of coordinate geometry and rotation from Lessons 1–3. Documenting predictions publicly creates a productive cognitive dissonance when the observed result differs, motivating deeper engagement with the construction activity.	Observable indicator: every student has labelled at least one predicted image vertex and written a justification sentence. Teacher scans for students who reflect instead of rotate (most common misconception) — these students are flagged for targeted questioning during the Observe phase. Note the range of predictions on the sticky note to revisit in the Explain phase.
Observe Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students work in groups of 3. Each group receives: a large sheet of squared paper (A3), a protractor, a ruler, and a pencil. Task Card A instructs them to draw triangle KMN (K(1,2), M(3,2), N(3,5)) on the Cartesian plane and rotate it 90° anticlockwise about the origin using the protractor-and-ruler method (draw a line from O to K, measure 90° anticlockwise, mark K′ at the same distance). They repeat for M and N, join the image vertices, and record coordinates in a table: Vertex | Object coords | Image coords. Task Card B then instructs the group to rotate quadrilateral ABCD (A(1,1), B(4,1), C(4,3), D(1,3) — representing the rectangular base plate of the turbine hub) by 180° about the	Before releasing groups, say: 'Remember — positive angle means anticlockwise, like the turbine spinning in the wind. Negative angle means clockwise. A 90° rotation is one quarter of a full turn around the hub.' Demonstrate ONE vertex rotation on the board (rotate point O-to-K line, measure 90° anticlockwise with protractor, mark K′ at same distance) — spend no more than 3 minutes on this. Say: 'Now it is your group's turn for the remaining vertices and all three Task Cards.' Circulate every 5 minutes. At each group stop: (1) ask 'How did you decide which direction is anticlockwise?' (WAIT 10 seconds before accepting answer); (2) check that students are measuring	Hands-On Coordinate Construction followed by Digital Verification — the physical construction with protractor and ruler builds procedural fluency and proprioceptive understanding of angle direction. The GeoGebra check immediately afterwards creates a feedback loop: students discover whether their manual technique was accurate, and the digital tool becomes evidence rather than a shortcut. Recording coordinates in a structured table primes students for the pattern-spotting needed in the Explain phase.	Observable indicators: (1) all three vertices of the image are equidistant from the centre of rotation as the corresponding object vertices (students can verify by measuring with ruler); (2) the completed coordinate table has correct image coordinates for at least Task Cards A and B; (3) GeoGebra output matches paper construction to within 1 unit. Teacher checks: ask any student to point to the centre of rotation and explain why it does not move — if they hesitate, this signals incomplete understanding of the invariant centre concept.

	origin, recording coordinates. Task Card C asks: rotate triangle PQR (P(2,0), Q(5,0), R(5,4)) by 90° clockwise (i.e. −90°) about the point (1,0). Groups measure side lengths of object and image with a ruler and record whether they are equal. After manual construction, one device per group is used to open GeoGebra (or the ARES offline GeoGebra module) to input the same shapes, apply the same rotations, and compare their paper coordinates against the digital output.	distance from the CENTRE, not from any other point; (3) for groups on Task Card C, prompt: 'The centre is not the origin this time — how does that change your first step?' Cold-call one member of each group on the image coordinates to check for arithmetic errors. If a group finishes early, extension: 'Rotate triangle KMN by 270° anticlockwise. What do you notice about the image compared to the 90° clockwise result?'		
Explain Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group shares their coordinate table from Task Cards A and B with the class (written on the board or displayed via a document camera). The class examines all tables together and is asked: 'Look at the object coordinates and the image coordinates for the 90° anticlockwise rotation — can you spot a pattern?' Students work in pairs for 3 minutes to write a conjecture. Groups share conjectures; the class co-constructs the coordinate rules: 90° anticlockwise: (x, y) → (−y, x); 180°: (x, y) → (−x, −y); 90° clockwise / 270° anticlockwise: (x, y) → (y, −x). Students then verify these rules against the GeoGebra outputs from Task Card C. Finally, students are asked: 'Return to the Ngong Hills turbine — if a blade tip is at position (4, 1) on our scale diagram, where will it be after one-third of a full turn anticlockwise (i.e. 120°)? Discuss with your partner.' This open question is intentionally not fully resolved — it plants the seed for Lesson 5 (rotation by non-standard angles). Students retrieve their folded	Write the two columns 'Object (x, y)' and 'Image (x′, y′)' for 90° anticlockwise on the board using values from the class tables. Say: 'Stare at these pairs for 30 seconds — what operation links x to x′, and y to y′?' WAIT 15 seconds in complete silence after posing the question. Cold-call 4 students before accepting any answer. When the pattern emerges, ask: 'Why does the negative sign appear with the x-coordinate in the image? Connect it to the direction of rotation.' Guide students to articulate: 'The original x-distance becomes the new downward distance, and the original y-distance becomes the new rightward distance.' Write the three rules in a box. Say: 'Now open your prediction sheet. Were you right? What was the most common mistake in our predictions?' Point to the board sketches from Phase 1. Validate the prediction effort: 'Making a careful prediction — even a wrong one — is exactly what scientists and engineers at companies like KenGen do before testing a new turbine design.' Close with the	Pattern Mining from Data Tables — students induce coordinate transformation rules from their own empirical data rather than receiving them as given formulae. This inductive approach embeds the rule in the context of the construction experience. The prediction-sheet revisit uses Structured Metacognitive Reflection to make the learning process visible and to address the most common misconceptions (confusion of rotation with reflection) directly and explicitly.	Observable indicators: (1) at least 3 out of 4 cold-called students can state the 90° anticlockwise rule correctly; (2) when asked to apply the rule to a new point — e.g. 'Use the rule to find the image of (2, −3) under 180° rotation about the origin' — at least 80% of students produce the correct answer (−(2), −(−3)) = (−2, 3) on their mini-whiteboards or in their exercise books; (3) students' prediction-sheet annotations show explicit identification of their own error type.

	prediction sheets from Phase 1, compare their original sketches with their constructed images, and annotate: 'I was right about… / I was wrong about… because…'	turbine application question and WAIT 12 seconds before taking responses.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class DQB (a large chart or shared digital board) displays the original driving question and sub-questions gathered in previous lessons. Each student writes on a sticky note one piece of NEW evidence gathered today that helps answer either: 'What stays the same during a rotation?' or 'How do we predict exactly where a point on the blade will land?' Students place sticky notes in the relevant column. The class then reads the board as a group and identifies: (1) questions that are now fully answered by today's lesson; (2) questions that are partially answered; (3) new questions that today's lesson has raised (e.g. 'What happens when the angle is not a multiple of 90°?'). The teacher records new questions in a different colour to signal that they will be addressed in future lessons.	Say: 'Today we rotated whole shapes — triangles and quadrilaterals — just like all the points on a turbine blade rotating together about the hub. Take 2 minutes to write your sticky note. Write one piece of evidence — something you observed or measured today — that helps answer our driving question.' WAIT while students write. After placing sticky notes, read 3–4 aloud and ask the class: 'Does this evidence help us answer the driving question? How?' Cold-call 2 students to connect the coordinate rules to the turbine context: 'Taraji, if the blade tip is at (3, 0) and the turbine does a quarter turn anticlockwise, where is the tip now — and how does the rule tell you that?' After the DQB update, say: 'We still cannot fully explain what happens at 120° — that is the puzzle we carry into the next lesson.'	Driving Question Board Evidence Sort — categorising sticky notes into 'answered', 'partially answered', and 'new questions' trains students to monitor the boundary of their own understanding. This metacognitive move is central to the NGSS practice of Obtaining, Evaluating, and Communicating Information, and keeps the anchoring phenomenon as the organising frame across all ten lessons.	Observable indicator: each student's sticky note contains a specific, measurable claim (e.g. 'The side lengths of the image triangle equal the side lengths of the object triangle — I measured them') rather than a vague statement (e.g. 'rotation is cool'). Teacher checks that the two driving sub-questions each receive at least one substantive evidence note from today's lesson.
Model Building Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of	Students retrieve their cumulative model — a scale diagram of the Ngong Hills turbine hub with three blades, started in Lesson 1 and updated each lesson. Today's task: (1) choose one blade (represented as a triangle on their model's coordinate overlay) and draw its image after a 120° anticlockwise rotation about the hub centre (the origin); (2) add congruence marks (tick marks) to matching sides of the object blade and its image to show that rotation preserves length; (3) add a curved	Say: 'Update your turbine model. A real KenGen wind turbine blade sweeps 120° between each blade position — that is one-third of a full 360° turn. Show this on your diagram with a rotation arc and congruence marks.' Circulate and ask targeted questions: 'Wanjiku, why did you put the tick marks on those two sides?' (WAIT 10 seconds). 'Samuel, your caption says the shape is unchanged — what measurement evidence from today supports that claim?' With 5 minutes remaining, ask two	Cumulative Annotated Model Revision — each lesson's model update adds a new layer of mathematical precision (today: congruence marks, rotation arcs, captions). This iterative diagrammatic modelling reflects the NGSS practice of Developing and Using Models and ensures that the anchoring phenomenon is continuously re-examined with growing sophistication. The caption-writing task requires students to integrate procedural knowledge (construction) with	Observable indicators: (1) the model shows a correctly positioned image blade with a curved arc labelled with the correct angle and direction; (2) congruence marks are placed on at least two pairs of matching sides; (3) the written caption uses the word 'congruent' (or 'same shape and size') and references the hub as the centre of rotation. Exit-pass check: as students leave, each shows the teacher their caption sentence — teacher gives a thumbs-up (sufficient) or

Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	rotation arc with an arrowhead and label it '120° anticlockwise'; (4) write a caption below the model: 'After rotating by 120° anticlockwise, Blade 1 maps exactly onto the position of Blade 2 — the shape and size are unchanged.' Students who finish early add the 240° rotation showing Blade 1 mapping to Blade 3, and annotate both arcs. The updated model is stored in the student's ARES portfolio for use in subsequent lessons.	volunteers to hold up their models and explain one new annotation to the class. Close the lesson by returning to the phenomenon: 'We started today asking why the turbine blades always look the same no matter where they stop. What can you now say about that — in one sentence, using the word congruent?' Cold-call 3 students for closing responses.	conceptual understanding (congruence preservation).	asks one clarifying question (needs revision before Lesson 5).

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Were students able to correctly apply the protractor-and-ruler method to rotate all vertices of a shape, or did most groups struggle with the non-origin centre in Task Card C? If Task Card C caused significant difficulty, plan a 10-minute warm-up at the start of Lesson 5 to revisit translation of the centre before rotation. (2) Did students induce the coordinate rules independently, or did the class need heavy scaffolding? If fewer than half the groups spotted the (x, y) → (−y, x) pattern without prompting, consider adding an intermediate column to the table in future (showing 'what happened to x?' and 'what happened to y?' as separate columns). (3) Were the DQB sticky notes specific and evidence-based? Vague notes indicate students have not yet internalised the evidence-gathering purpose of the lesson sequence — reinforce this norm explicitly at the start of Lesson 5. (4) Did the model-building phase connect meaningfully to the 120° blade spacing of a real wind turbine, or did it feel artificially bolted on? If the connection felt weak, bring in a clearer image or short video clip of the Ngong Hills turbine at the start of Lesson 5 to re-anchor the phenomenon before the lesson proceeds.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that when every vertex of triangle KMN was rotated 90° anticlockwise about the origin, the resulting image triangle K′M′N′ had exactly the same side lengths and angles as the original — confirmed by measurement with a ruler and by GeoGebra. They also observed that rotating quadrilateral ABCD by 180° about the origin produced an image that was congruent to the object and positioned symmetrically about the origin.
What did I learn?	Students learned the coordinate transformation rules for rotation about the origin: 90° anticlockwise maps (x, y) to (−y, x); 180° maps (x, y) to (−x, −y); 90° clockwise maps (x, y) to (y, −x). They also learned that rotation about a non-origin centre requires each vertex to be rotated through the same angle about that specific centre point, and that positive angles indicate anticlockwise direction while negative angles indicate clockwise direction.
How does this explain the phenomenon?	Students explained that the Ngong Hills turbine blades always look the same no matter where they stop because rotation is a congruence-preserving transformation — every point on the blade moves through the same angle about the fixed hub, so the overall shape and size of the blade is unchanged. The three blades, spaced 120° apart, each map exactly onto the next blade's position after a one-third turn, which is why the turbine assembly appears identical at every stopping point.

LESSON 5 (80 minutes): Angle and Centre of Rotation — Finding What We Cannot See	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to determine the angle and centre of rotation given an object and its image using geometric construction.
Knowledge	Learners will know that the centre of rotation is equidistant from each object point and its corresponding image point, and that perpendicular bisectors of segments joining corresponding points are concurrent at the centre of rotation.
Skills	Learners will be able to construct perpendicular bisectors of segments joining corresponding object-image point pairs to locate the centre of rotation, and use a protractor to measure the angle of rotation.
Attitudes	Learners will appreciate that logical geometric reasoning — not guesswork — is the reliable tool for uncovering hidden mathematical structure, and will value precision in construction.
Key Inquiry Question	How do we determine the centre and angle of rotation given an object and its image?
Purpose in Storyline	In Lessons 1–4 students established what rotation is, how direction and angle are defined, and what properties are preserved. Now, in Lesson 5, students flip the challenge: instead of performing a rotation they must reverse-engineer one. Given only a blade position and its rotated image, can they mathematically locate the hidden centre — the turbine hub — using construction alone? This resolves a critical piece of the driving question: the hub's fixed position is not merely visible; it is geometrically recoverable from any two blade positions.
Safety Notes	Ensure compasses are handled with care — tips are sharp. Students should place compasses on the desk when not in use. Remind learners not to swing rulers or compasses during pair work.

B. LESSON OVERVIEW

Learners receive printed object-image pairs representing a single turbine blade in two positions. Using ruler, compass, and protractor, they construct perpendicular bisectors of segments connecting corresponding vertices to locate the centre of rotation, then measure the angle of rotation. A short reading on logical reasoning in geometry scaffolds the sensemaking. The lesson culminates with students applying their construction to the Ngong Hills turbine context: if one blade sweeps to a new position, the construction reveals exactly where the hub must be. Learners update their Driving Question Board and revise their cumulative model of rotation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Proofs with transformations] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos]	Learners are shown two printed diagrams side by side on their activity sheets: Diagram A shows a single turbine blade (a thin triangle) labelled with vertices P, Q, R. Diagram B shows its image P', Q', R' in a different orientation on the same grid. No centre is	Distribute activity sheets face-down. Say: 'Do not flip until I say go.' Reveal both diagrams on the board using the projected image. Say: 'You have seen us rotate shapes in previous lessons. Now the rotation has already happened — your job is to find the hidden	Prediction Mapping — learners externalise prior intuition about rotational symmetry before instruction. This creates cognitive dissonance when construction reveals the precise answer, motivating engagement with the construction procedure as a	Observe whether learners place their predicted centre near the midpoint region between object and image (common misconception) versus near the origin or another logically justified location. Listen for language such as 'equal distance', 'middle', or

📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	marked. Learners are asked: 'Without measuring anything yet — where do you think the centre of rotation is? Mark it with an X on your sheet and write one sentence explaining your reasoning.' Learners work individually for 3 minutes, then share predictions with a shoulder partner.	centre. Mark your best prediction with an X and write ONE sentence: why there?' Give 3 minutes of silent individual thinking. After 3 minutes say: 'Turn and share with your partner. You have 90 seconds.' Cold-call 3 pairs — ask: 'Where did you place your X and what was your reasoning?' Do NOT confirm or deny predictions. Record 4–5 diverse predictions on the board with sticky dots, labelled 'Our Predictions'. Say: 'By the end of this lesson, construction — not guessing — will tell us exactly who is closest.'	resolution tool.	'same as before' — this reveals whether learners are drawing on distance-preservation from Lesson 4 or relying on pure visual guessing.
Observe Phase 📹 **VIDEO:** [Proofs with transformations] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners read a one-page excerpt titled 'Reasoning Backwards in Geometry' (provided on the activity sheet). The reading explains: (1) In geometry, if a transformation has occurred, the properties it preserves leave behind clues. (2) Because rotation preserves distance from the centre, every object point and its image are equidistant from the centre — so the centre lies on the perpendicular bisector of the segment joining each object-image pair. (3) Two perpendicular bisectors from two different pairs will intersect at exactly one point: the centre. After reading, learners watch a 4-minute teacher-led live demonstration on the board: the teacher constructs the perpendicular bisector of segment PP' (compass arcs above and below, ruling the bisector), then constructs the perpendicular bisector of segment QQ', and marks the intersection. Learners observe silently and annotate their reading with arrows and labels as the demonstration proceeds.	Say: 'Before we construct anything, we read to understand WHY the method works — not just HOW.' Allow 5 minutes of silent reading. Then say: 'Underline one sentence in the reading that explains WHY the centre lies on a perpendicular bisector. You have 1 minute.' Cold-call 2 learners to read their underlined sentence aloud. Wait 10 seconds after each. Then move to the board: 'Watch carefully — I will construct this once, slowly. Your job is to annotate your diagram as I go.' Narrate every step aloud: 'I place my compass on P — wider than half of PP' — I draw an arc above and below the line. Same width, now on P'. Where the arcs cross, I draw the bisector. This line contains all points equidistant from P and P' — so if our centre is equidistant from P and P', it lives on this line.' Repeat for QQ'. Mark intersection O. Say: 'This point O is the only point equidistant from both pairs. That is our centre.' Pause 15 seconds. Ask: 'What do you notice about where O landed compared to your prediction?'	Read-then-Watch Sequencing — learners first encounter the logical justification in text, then see it enacted visually. This dual-coding approach (verbal reasoning + geometric construction) ensures learners understand the 'why' before attempting the 'how', reducing the risk of procedural mimicry without understanding.	Check that learners underline a sentence referencing equidistance or perpendicular bisector (not a sentence about procedure). During the demonstration, circulate and check that learners are annotating — specifically that they label the bisector lines and the intersection point O. A learner who annotates correctly but cannot explain why the bisectors intersect at O will need targeted questioning in Phase 3.

		Cold-call 3 students.		
Explain Phase 📹 **VIDEO:** [Proofs with transformations] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners now independently construct the centre and angle of rotation for two object-image pairs on their activity sheet. Task 1: Triangle KLM and its image K'L'M' (a turbine blade rotated about an unmarked hub). Learners construct perpendicular bisectors of KK' and LL', mark the centre O, draw rays OK and OK', then use a protractor to measure the angle of rotation and state its direction (clockwise or anti-clockwise). Task 2: A single point J and its image J' given that the centre of rotation is already marked — learners must verify by constructing the perpendicular bisector and confirming it passes through the given centre, then state the angle. After completing both tasks, learners write a two-sentence explanation: 'The centre of rotation is found by… The angle of rotation is measured by…' in their exercise books. Pairs then compare constructions and resolve any discrepancies by reasoning, not by copying.	Say: 'You will now do what I demonstrated — but on your own sheet. Read each task instruction fully before drawing anything.' Circulate during Task 1. At the 7-minute mark, pause the class: 'Hold up your pencil if you have found your centre O.' Scan the room. For learners who have not, say quietly: 'Construct the bisector of KK' first — does it cross your paper? Good. Now QQ'. Where do they meet?' Do NOT draw on their sheets. After Task 2, ask the whole class: 'Did the perpendicular bisector of JJ' pass through the given centre?' Wait 10 seconds. Cold-call 3 learners — ask them to state YES or NO and explain WHY in one sentence. Then prompt: 'Mwangi, what angle did you measure for Task 1, and is the rotation clockwise or anti-clockwise?' Repeat for Amina and Kipchoge. Write all three answers on the board. If there is disagreement, say: 'Let us reason through this together — a positive angle means anti-clockwise, negative means clockwise. Look at your diagram: which way did the blade sweep from K to K' around centre O?'	Worked-Example to Independent Practice Transfer — after observing the teacher model (Phase 2), learners execute the same process independently with a new figure. The two-sentence written explanation forces metacognitive articulation: learners must translate procedural steps into conceptual language, consolidating understanding of WHY the method works.	Collect and scan 6 activity sheets mid-task. Look for: (1) Are both perpendicular bisectors drawn (not just one)? (2) Is the intersection clearly labelled O? (3) Is the angle measured from ray OK to ray OK' at the centre — not at a vertex of the triangle? (4) Does the direction (CW/ACW) match the diagram? Common error: learners measure the angle at K instead of at O — address this whole-class if more than 3 learners make this error.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners return to the class Driving Question Board. The board has two anchor questions: (1) 'Why do the blades always look the same no matter how far they rotate?' and (2) 'How can we predict exactly where any point on the blade will land?' Learners receive two sticky notes each. On the first (green): they write one new thing they can NOW answer or partially answer using today's	Point to the driving question on the board and read it aloud: 'If we know one blade position and its rotated image, can we find the centre?' Say: 'Three lessons ago you could not have answered this. Today, write on your green sticky note: what CAN you now say in answer to this question?' Give 2 minutes of silent writing. Then: 'On your orange note, write what is still puzzling or what you want to know	Driving Question Board (DQB) Protocol — a structured public record of growing understanding. By distinguishing 'what I can now explain' from 'what I still wonder', learners practise scientific epistemic humility and track their own conceptual progress across the 10-lesson sequence. Orange notes feed directly into lesson planning for Lessons 6–10.	Read all green sticky notes after class. A proficient response will reference perpendicular bisectors, equidistance, or construction as the method for finding the centre. A developing response will describe the visual location without referencing the geometric justification. Learners who write only 'I found the centre by drawing lines' need follow-up questioning in Lesson 6 to probe their

📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	learning. On the second (orange): they write one question that today's lesson has raised or left unanswered. Sticky notes are placed in the appropriate columns on the DQB. The class does a 3-minute gallery walk to read peers' contributions before the teacher facilitates a 4-minute whole-class discussion connecting today's construction to the turbine phenomenon.	next.' Collect and place notes on the DQB during the gallery walk. After the gallery walk, cold-call 2 green notes and 2 orange notes — read them aloud without naming the author. Respond to one orange note by saying: 'This is a great unresolved question — let us note it here. Our next lessons will address this.' Explicitly state: 'Today we learned that the hub of the Ngong Hills turbine is not just a visible bolt — it is a geometrically necessary point. Any two blade positions contain enough information to reconstruct exactly where it must be.'		understanding of WHY the construction works.
Model Building Phase 📹 **VIDEO:** [Proofs with transformations] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative rotation model — a labelled diagram they have been building since Lesson 1 to explain the Ngong Hills turbine phenomenon. Today they add two new annotated elements: (1) They draw a two-blade configuration (object blade and image blade) and construct the perpendicular bisectors of two corresponding point pairs, labelling the intersection as the centre of rotation O (the turbine hub). They annotate: 'The hub is the unique point equidistant from every object point and its corresponding image.' (2) They draw an angle arc at O between the two corresponding rays and label it with the angle of rotation and its direction. Below the diagram, learners write a 'Model Claim': 'Given any two positions of a turbine blade, I can find the hub and the angle swept using perpendicular bisector construction.' Learners share their updated model with a partner and give one piece of specific feedback using the stem: 'Your model	Say: 'Take out your cumulative model. You are about to add the most powerful piece yet — the ability to find the hidden centre from evidence alone.' Display the model-building prompt on the board: '(1) Draw object and image blade. (2) Construct perpendicular bisectors of two point pairs. (3) Label centre O — the hub. (4) Mark and label the angle of rotation. (5) Write your Model Claim.' Give 10 minutes for silent model revision. Circulate and look for: Is the angle drawn at O (not at a vertex)? Is the construction visible — or did they just mark a dot? Prompt where needed: 'Show me the construction lines — they are evidence of your reasoning.' After 10 minutes: 'Share with your partner. Give feedback using the sentence starter on the board.' Give 3 minutes for peer feedback. Close by cold-calling one learner to hold up their model and narrate it in two sentences to the class. Ask the class: 'Does this model now help us answer our driving question? What is still missing?'	Cumulative Model Revision — learners do not create a new diagram each lesson; they add to one growing artefact. This mirrors how scientists build and refine models as evidence accumulates. The 'Model Claim' sentence forces learners to state what their model now allows them to predict — linking construction skill to explanatory power over the turbine phenomenon.	Collect 5 models at random at the end of class. Evaluate against three criteria: (1) Are perpendicular bisector construction arcs visible (not just a ruled line)? (2) Is the angle of rotation measured and labelled at centre O with direction stated? (3) Does the Model Claim connect the construction to the turbine hub specifically — not just a generic statement about rotation? Models meeting all three criteria are proficient. Models missing criterion 3 indicate that learners can execute the procedure but have not yet connected it back to the anchoring phenomenon.

	shows… but it could be stronger if you also showed…'	Wait 15 seconds before taking responses.		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) How many learners correctly identified the centre of rotation using perpendicular bisectors without teacher intervention? If fewer than 70% succeeded independently, consider opening Lesson 6 with a paired re-do of Task 1 using a fresh figure. (2) Did learners measure the angle of rotation at centre O, or at the object vertices? If the latter error was widespread, build a 5-minute targeted correction into the start of Lesson 6 using the prompt: 'The angle of rotation is always the angle swept AT THE CENTRE — think of standing at the turbine hub and watching the blade sweep past you.' (3) Review the orange sticky notes from the DQB. Are learners asking questions about multiple rotations, or about what happens when there is only one corresponding pair given? These questions should directly inform the entry activity of Lesson 6. (4) Did the reading excerpt successfully build the logical justification, or did learners skip to the procedure? If annotations were sparse, consider replacing the silent reading in Lesson 6 with a teacher-facilitated Think-Aloud of a similar text. (5) Note which learners produced strong Model Claims connecting construction to the turbine hub — these learners can serve as peer explainers in Lesson 6's opening review.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed a teacher-led compass-and-ruler construction showing how perpendicular bisectors of segments joining object-image point pairs (PP' and QQ') intersect at a single point — the centre of rotation. They also read an explanation of WHY this works: because rotation preserves distance from the centre, every object-image pair is equidistant from the centre, so the centre must lie on the perpendicular bisector of each such segment.
What did I learn?	Learners learned that: (1) the centre of rotation is the unique point equidistant from every object point and its corresponding image point; (2) constructing perpendicular bisectors of two object-image point segments and finding their intersection locates the centre; (3) the angle of rotation is measured at the centre O between the ray to the object point and the ray to the image point; (4) a positive (anti-clockwise) or negative (clockwise) sign must accompany the angle. They also learned that this construction can reverse-engineer a rotation — given only two blade positions, the turbine hub's location is fully determined.
How does this explain the phenomenon?	Learners explained how — given only the positions of a Ngong Hills turbine blade before and after a rotation — they can construct perpendicular bisectors of two corresponding vertex segments to locate the exact centre of rotation (the turbine hub), and then use a protractor at that centre to measure and state the angle and direction of rotation. They updated their cumulative model and DQB to reflect that the hub is not merely a visible physical point but a geometrically necessary and recoverable centre of rotational symmetry.

LESSON 6 (80 minutes): Order of Rotational Symmetry	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to investigate and define the order of rotational symmetry for regular polygons and real-world Kenyan patterns, connecting this property to the Ngong Hills wind turbine phenomenon.
Knowledge	By the end of the lesson, the learner should be able to: (a) define order of rotational symmetry; (b) determine the order of rotational symmetry for regular polygons (equilateral triangle, square, regular hexagon) and Kenyan cultural patterns (Maasai beadwork, Kanga tiling designs); (c) calculate the minimum angle of rotation that maps a shape onto itself using the formula 360° ÷ order; (d) explain why the three-bladed Ngong Hills turbine has rotational symmetry of order 3.
Skills	Identifying lines/angles of rotational symmetry; applying the formula 360° ÷ n to compute minimum rotation angle; classifying shapes by their order of rotational symmetry; constructing and revising explanatory models.
Attitudes	Appreciate the presence of rotational symmetry in Kenyan cultural art and engineering design; demonstrate curiosity and persistence when investigating patterns; value collaborative discussion in building mathematical understanding.
Key Inquiry Question	1. What are the properties of rotation as a geometric transformation? 2. How do we determine the centre and angle of rotation given an object and its image?
Purpose in Storyline	In Lessons 1–5 students established the mechanics of rotation: direction (positive = anticlockwise, negative = clockwise), angle measurement, centre of rotation, and image coordinates. Lesson 6 now uses that foundation to answer the anchoring puzzle directly — why does the turbine blade assembly always look the same? Students discover that a shape has rotational symmetry of a specific order when it maps onto itself at equal fractional turns, and they quantify exactly which angles produce those self-mappings. This is the conceptual turning point of the unit: students move from 'how do we rotate a point?' to 'why does a rotating object look unchanged?'
Safety Notes	No physical hazards anticipated. Ensure screen brightness is comfortable during video viewing. If printed Maasai beadwork/Kanga pattern cards are used, handle with care to avoid paper cuts. Remind students to respect and appreciate the cultural significance of Maasai and Kanga designs when discussing them mathematically.

B. LESSON OVERVIEW

Learners begin by revisiting the Ngong Hills turbine phenomenon and predicting how many times the three-bladed assembly maps onto itself in one full turn. They then watch a guided video that systematically rotates regular polygons and cultural Kenyan patterns about their centres, recording each angle at which the shape looks identical to the original. Through structured group sensemaking, learners define order of rotational symmetry, derive the formula minimum angle = 360° ÷ order, and apply it to classify several shapes including Maasai beadwork rosettes and square Kanga tiling blocks. The lesson culminates in a model revision where learners annotate their running turbine diagram with the order of symmetry and the exact rotation angles at which each blade returns to an equivalent position. The Driving Question Board is updated with new, more precise answers and refined questions about symmetry in nature and design.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Even and Odd Functions (Symmetry) (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners examine a still photograph of the Ngong Hills three-bladed turbine displayed on screen. Working individually in their exercise books, they respond to two written prompts: (1) 'If the turbine rotates anticlockwise, list ALL the angles between 0° and 360° at which the blade assembly looks exactly the same as it does right now. Explain your reasoning.' (2) 'A Maasai beadwork rosette has 6 equal petal sections. How many times would you expect it to look the same in one full turn? What angles would those be?' After 4 minutes of silent individual writing, learners share predictions with a shoulder partner for 2 minutes. Three pairs are cold-called to share their reasoning with the class.	Display the turbine photograph and read both prompts aloud. Say: 'You have four minutes to write your predictions silently — use what we have already learned about rotation angles and direction.' [Allow 4 minutes silent work.] After partner sharing, cold-call exactly 3 pairs: 'Pair at the front left — what angles did you predict for the turbine, and why?' [WAIT TIME: 12 seconds.] 'Pair in the middle row — do you agree or did you predict differently?' [WAIT TIME: 10 seconds.] 'Pair at the back right — what about the Maasai rosette?' [WAIT TIME: 12 seconds.] Record predicted angles on the board in two columns: TURBINE | MAASAI ROSETTE. Do NOT confirm or deny any prediction. Say: 'Keep these predictions in mind — the video we are about to watch will give us evidence to test every single one of them.'	Individual written prediction followed by Think-Pair-Share. This activates prior knowledge of rotation angles (Lessons 1–5) and creates a public record of pre-instruction thinking. By making predictions concrete and visible on the board, cognitive conflict is primed — learners are motivated to watch the video carefully to verify or refute their own ideas.	Teacher circulates during silent writing and notes: (a) Do learners attempt to space angles equally (evidence of emerging symmetry intuition)? (b) Do learners correctly use 360° as the full turn reference? (c) Do any learners write negative angles, demonstrating recall of clockwise convention? These observations inform the level of scaffolding needed during the Observe phase.
Observe Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Even and Odd Functions (Symmetry) (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners watch a structured 12-minute ARES video titled 'Symmetry in Spin: From Ngong Hills to Maasai Patterns.' The video opens with slow-motion footage of the Ngong Hills turbine rotating, with a dot marked on one blade tip. An overlay shows the rotation angle increasing from 0°. The video pauses at 120°, 240°, and 360° to show that the blade assembly looks identical each time. The video then transitions to animated regular polygons (equilateral triangle, square, regular pentagon, regular hexagon) rotating about their centres, each time pausing and highlighting when the shape maps onto itself. Next, the video shows two Kenyan cultural designs: a	Before playing the video, distribute (or display) the observation table template. Say: 'As you watch, fill in your observation table for every shape shown. Pause your thinking at each freeze-frame — the video will give you time to write.' Play the video without interruption. After the video ends, say: 'Now compare your completed table with the person next to you. Do your numbers match? If not, discuss why.' [Allow 3 minutes pair discussion.] Cold-call 2 students: 'Tell me — for the equilateral triangle, which angles did you record?' [WAIT TIME: 10 seconds.] 'For the regular hexagon, how many times did it map onto itself?' [WAIT TIME: 10 seconds.] Write confirmed class values on the	Structured observation table with a Claim-Evidence link. The table format forces learners to record specific numerical evidence (angles and counts) rather than vague impressions. Comparing individual tables in pairs immediately after watching promotes collaborative accuracy checking and surfaces discrepancies for class discussion, deepening engagement with the data.	Teacher scans completed observation tables during pair discussion. Look for: (a) Correct listing of equally spaced angles for each shape (e.g., 120°, 240°, 360° for the triangle); (b) Accurate counts (equilateral triangle → 3, square → 4, regular hexagon → 6); (c) Whether learners include 360° in their list — a common point of contention that will be addressed in Phase 3. Flag students who list only the non-360° angles for targeted questioning in Explain.

	Maasai beadwork circular rosette (6-petal) and a square Kanga tile border unit. Finally, the video poses: 'How many times does each shape fit onto itself in one full 360° turn? What is the smallest angle that makes this happen?' Learners complete an observation table in their books with columns: Shape | Angles at which it maps onto itself | Number of times it maps onto itself in 360°.	board next to the predicted values from Phase 1. Ask: 'Were any of your predictions from Phase 1 correct? Were any surprising?' [WAIT TIME: 12 seconds — take 2–3 hands.]		
Explain Phase 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Even and Odd Functions (Symmetry) (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four to complete a structured sensemaking task. Each group receives a set of shape cards (equilateral triangle, square, regular pentagon, regular hexagon, a Maasai 6-petal rosette, a square Kanga tile unit, and the Ngong Hills 3-blade turbine outline). Groups: (1) Write a definition of 'order of rotational symmetry' in their own words using evidence from their observation tables. (2) Complete a class results table on their paper: Shape | Order of Rotational Symmetry | Minimum Angle of Rotation. (3) Identify the pattern between the order and the minimum angle, and write it as a formula. (4) Answer the focus question: 'A new Ngong Hills turbine design has 4 blades spaced equally. What is its order of rotational symmetry and what are ALL the angles at which it maps onto itself?' Groups then share findings. The teacher leads a whole-class discussion to formalise the definition and formula: Minimum Angle = 360° ÷ Order of Rotational Symmetry.	Assign groups and distribute shape cards. Say: 'Your group has 12 minutes to complete all four tasks on the sheet. Use your observation table as your evidence — you should not need to guess anything.' [Circulate, checking group progress. At 6 minutes, prompt any group still on task 1: 'You have completed the table — what pattern do you see between the order and the minimum angle?'] After group work, select 2 groups to present task 3 (the formula). Ask: 'Group by the window — what formula did your group write? Where did the 360 come from?' [WAIT TIME: 12 seconds.] After both groups present, say: 'Let us now write the formal mathematical definition together.' Write on the board: 'A shape has ROTATIONAL SYMMETRY OF ORDER n if it maps onto itself exactly n times during a full 360° rotation. The minimum angle of rotation = 360° ÷ n.' Ask: 'Using this formula, what is the minimum angle for the three-bladed Ngong Hills turbine?' [WAIT TIME: 10 seconds.] Cold-call 1 student. Then ask: 'So does this explain why the turbine always looks the same? Turn and tell your partner in ONE sentence.' [WAIT	Collaborative Group Inquiry followed by Class Formalisation (Gradual Release of Authority). Groups construct the definition and formula inductively from their own evidence before the teacher formalises it. This ensures learners experience the formula as something they derived, not something imposed, which builds durable conceptual understanding. Connecting the formula back to the turbine immediately contextualises the abstract result.	During group work, listen for: (a) Can groups articulate the definition of order without prompting? (b) Do groups correctly compute minimum angles (e.g., pentagon: 360°÷5 = 72°)? (c) Can groups extend to the 4-blade turbine independently? During whole-class sharing, note whether students can explain WHY 360° is divided — understanding that 360° is one full turn. Students who compute the formula correctly but cannot explain the WHY are flagged for targeted questioning or a follow-up exit card.

		TIME: 15 seconds — take 3 responses.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Even and Odd Functions (Symmetry) (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front of the room. Individually, each learner writes on a sticky note (or in their books if sticky notes are unavailable) a specific updated answer to the driving question: 'Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate?' They must use the phrase 'order of rotational symmetry' and cite at least one specific angle in their response. Learners also add a new 'refined question' sticky note — something they are now curious about that they were not thinking about at the start of the lesson (e.g., 'Does the number of blades affect how much electricity the turbine generates?' or 'Do all natural shapes with rotational symmetry have a formula like this?'). The teacher reads selected notes aloud and places them in the ANSWERS column and the NEW QUESTIONS column of the DQB.	Direct attention to the DQB. Say: 'In Lesson 1 we asked why the turbine always looks the same. You now have mathematical language to answer that. Write your best answer on your sticky note — it must include the phrase ORDER OF ROTATIONAL SYMMETRY and at least one angle.' [Allow 3 minutes writing.] Collect 5–6 answer notes and read them aloud, paraphrasing for the class. Say: 'Listen to this answer — it says the turbine has order 3 because 360°÷3 = 120°, so every 120° anticlockwise the assembly looks identical. Does everyone agree? Is there anything missing from this answer?' [WAIT TIME: 12 seconds — take 2 responses.] Pin the strongest answer to the DQB ANSWERS column. Then say: 'Now write a new question — something this lesson has made you wonder about.' Pin 3–4 new question notes to the NEW QUESTIONS column, reading each aloud. Say: 'These new questions will guide Lessons 7 to 10. Hold onto your curiosity.'	Driving Question Board Cumulative Update. The DQB makes the intellectual journey of the unit visible and tangible. By requiring learners to use specific mathematical vocabulary in their answers, the update consolidates new terminology in a meaningful context. The new questions column honours learner curiosity and signals that science and mathematics are ongoing, generative processes — not closed answer-giving exercises.	Teacher reads all collected sticky notes (or scans the room's books) to assess: (a) Are learners correctly using 'order of rotational symmetry' with an accurate order for the turbine (order 3)? (b) Do learners cite 120° and 240° as the key angles? (c) Are new questions mathematically or scientifically substantive? Notes that contain misconceptions (e.g., 'order 6 because there are 6 blade edges') are addressed immediately by comparing with the observation table evidence from Phase 2.
Model Building Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Even and Odd	Learners open their running Ngong Hills turbine diagram — a cumulative annotated model they have been building since Lesson 1. Today's revision task: (1) Label the centre of rotation on the hub. (2) Draw three arcs from a marked blade tip showing the 120°, 240°, and 360° rotation positions, using a protractor to ensure accuracy, and label each arc with its angle. (3) Write inside the diagram: 'Order of rotational symmetry = 3. Minimum rotation angle = 120°.'	Say: 'Open your running turbine model. Today we are adding the most important annotation yet — the one that directly answers our driving question.' Display a partially completed model on the board or projector as a reference. Say: 'Use your protractor to draw the arcs accurately — the angles must be correct, not estimated.' Circulate and check protractor use. At 5 minutes, prompt: 'Have you written your sentence connecting the equal spacing of blades to the	Cumulative Annotated Model Revision (Progressive Diagrammatic Explanation). Each lesson's model revision adds a new layer of mathematical meaning to the same diagram, externalising the learner's growing understanding. Using a protractor to draw arcs accurately reinforces the precision of mathematical rotation and connects procedural skill (angle measurement) to conceptual understanding (self-mapping at exact angles). The	Teacher checks models for: (a) Accurate arc placement at exactly 120° and 240° from the marked blade tip (not estimated); (b) Correct labelling of order = 3 and minimum angle = 120°; (c) A coherent sentence that explicitly links 'three equally spaced blades' to 'order 3 symmetry' (not just restating the definition). Models with inaccurate arc angles indicate a gap in protractor skill or angle calculation that must be addressed before Lesson 7. Extension

Functions (Symmetry) (PLIX) Source: CK-12 🔍 Search ARES for similar readings	(4) Add a colour-coded key: green arc = 120° (first self-mapping), blue arc = 240° (second self-mapping), red arc = 360° (full turn, back to start). (5) Write one sentence below the diagram connecting the order of symmetry to the equal spacing of the three blades. Learners who finish early extend the model by sketching a 4-blade and 6-blade turbine variant and annotating each with its order and minimum angle.	order of symmetry? That sentence is the heart of the answer to our driving question.' At 8 minutes, cold-call 2 students to share their sentence with the class: 'Read your sentence aloud — does your sentence explain the WHY or just the WHAT?' [WAIT TIME: 10 seconds each.] Close the model activity by saying: 'In Lesson 7 we will use this model again when we explore how to find the centre of rotation when it is not given to us. Make sure your model is accurate and complete — it is a tool you will use all the way to Lesson 10.'	sentence-writing task integrates verbal and visual representations, strengthening retention.	models (4-blade and 6-blade turbines) are used to identify learners ready for enrichment tasks.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Could learners articulate the definition of order of rotational symmetry in their own words before I formalised it — or did most groups wait for me to provide the definition? If the latter, how can I reduce teacher authority earlier in the Explain phase in the next lesson? (2) Did learners successfully connect the abstract formula (360° ÷ n) back to the physical turbine blades, or did the formula feel disconnected from the phenomenon? If disconnected, consider opening Lesson 7 with a brief re-link to the turbine before introducing finding the centre of rotation. (3) Were the Maasai beadwork and Kanga tile examples culturally affirming and mathematically productive? Did learners engage with them seriously, or were they treated as decoration? Consider inviting learners to bring additional examples of rotational symmetry from their own homes or communities. (4) Review the sticky-note DQB answers: how many learners correctly stated order 3 with specific angles? Use this count as a percentage-correct indicator — if fewer than 70% are correct, re-teach the minimum angle formula at the start of Lesson 7 before advancing. (5) Check the accuracy of learners' protractor work in their model diagrams. Persistent inaccuracy in angle drawing will undermine Lesson 7 (finding the centre of rotation), so consider a 5-minute protractor drill as a warm-up activity next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students watched a video of the Ngong Hills three-bladed wind turbine and animated regular polygons (equilateral triangle, square, regular pentagon, regular hexagon) rotating about their centres, as well as a Maasai beadwork 6-petal rosette and a square Kanga tile unit. They recorded in an observation table every angle at which each shape mapped onto itself within one full 360° turn, and counted the number of times each shape did so.
What did I learn?	A shape has rotational symmetry of order n if it maps exactly onto itself n times during a full 360° rotation. The minimum angle of rotation that produces a self-mapping is calculated as 360° ÷ n. The Ngong Hills three-bladed turbine has rotational symmetry of order 3, with self-mappings at 120°, 240°, and 360°. This is why the blade assembly always looks the same at those positions — the rotation maps every blade onto the position of another blade.
How does this explain the phenomenon?	The three-bladed Ngong Hills turbine always looks the same at certain rotation positions because the three blades are equally spaced at 120° apart, giving the assembly a rotational symmetry of order 3. Every time the turbine rotates 120° anticlockwise (positive rotation), each blade moves into the exact position the previous blade occupied, so the overall shape is unchanged. This

	is a property called rotational symmetry, and the order tells us how many times this identical mapping occurs in one full 360° turn.

LESSON 7 (40 minutes): Axis and Order of Rotational Symmetry: From 2D Figures to 3D Objects	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will distinguish between the order of rotational symmetry for 2D figures and the axis of rotational symmetry for 3D objects, and apply both concepts to familiar Kenyan objects to consolidate their understanding of the turbine phenomenon.
Knowledge	Students will know that: (1) a 3D object has an axis of rotational symmetry if rotation about that axis maps the object onto itself; (2) the order of rotational symmetry for a 3D object about a given axis is determined in the same way as for 2D figures — the number of times the object maps onto itself in one full rotation; (3) everyday Kenyan objects such as a sufuria, a wheel, a gear, and the Ngong Hills turbine hub all possess rotational symmetry about a defined axis.
Skills	Students will be able to: (1) identify and describe the axis of rotational symmetry of given 3D objects; (2) determine the order of rotational symmetry of 2D and 3D objects; (3) distinguish between the axis of symmetry concept applied to 2D versus 3D contexts; (4) record and justify their findings using correct mathematical language.
Attitudes	Students will appreciate that mathematics is embedded in familiar everyday objects and that precise geometric language allows them to describe real-world phenomena clearly and confidently.
Key Inquiry Question	What are the properties of rotation as a geometric transformation? (extended to 3D: how does the concept of self-mapping under rotation apply when we add a third dimension and introduce an axis?)
Purpose in Storyline	Lesson 6 established that the three-bladed turbine has rotational symmetry of order 3 in 2D. Lesson 7 extends this to 3D by naming the axis about which the turbine rotates — the horizontal shaft through the hub — and by exploring other Kenyan 3D objects. This prepares students to make a complete, precise claim in Lesson 8 about congruence and in Lesson 10 about the full driving-question answer, which requires both order and axis language.
Safety Notes	No sharp or electrical objects to be handled. Physical objects (sufuria, bottle caps, gear cut-outs) should be clean and free of sharp edges. Students should not spin any object near their faces.

B. LESSON OVERVIEW
Students begin by recalling the order of rotational symmetry of the turbine (order 3, minimum angle 120 degrees) from Lesson 6. They are then challenged: the turbine does not just rotate on paper — it rotates in space about a real physical line called an axis. Through a predict-observe-explain cycle, students examine 3D objects (a sufuria, a circular Kenyan bread tin lid, a bottle cap, a drawn gear wheel) to identify axes of rotational symmetry and determine orders. They consolidate the 2D-versus-3D distinction and update the class Driving Question Board with the new precise language needed to fully answer the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Electrochemical	The teacher holds up a sufuria (cooking pot) and asks students to write silently for 90 seconds: (1) If I	Hold the sufuria at eye level so all students can see it clearly. Say: 'Before we touch anything or watch	Bridging from prior knowledge — students explicitly connect the Lesson 6 definition of order of	Listen for whether students correctly recall that order equals the number of self-mappings in

gradients and secondary active transport Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Even and Odd Functions (Symmetry) (PLIX) Source: CK-12 🔍 Search ARES for similar readings	spin this sufuria around an imaginary vertical line through the centre of its base, how many times will it look exactly the same before completing one full turn? (2) Is there any other line I could spin it around that would also give the same effect? Students sketch the sufuria, draw what they think the axis looks like, and write their predicted order. They share predictions with a partner before a brief whole-class share-out of two or three predictions.	anything, I want your best scientific guess. Think about what we learned in Lesson 6 about order of rotational symmetry — now imagine lifting that idea off the flat page and into this real 3D object.' Allow 90 seconds of silent individual thinking. WAIT TIME: after asking the two questions, say nothing for a full 90 seconds — resist the urge to prompt. After pair sharing, cold-call two students: 'Amina, what order did you predict and why?' and 'Brian, did you predict a different axis — describe it to us.' Record both predictions on the board without evaluating them.	rotational symmetry to a new context. Predictions are made visible on the board so they can be confirmed or revised in the Observe phase.	360 degrees. If students say the sufuria maps onto itself infinitely (order infinity), note this as a productive idea to revisit in the Explain phase — a circle is a special case. Check whether students can articulate what an axis is, even informally.
Observe Phase 📹 **VIDEO:** Electrochemical gradients and secondary active transport Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Even and Odd Functions (Symmetry) (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students work in groups of four. Each group receives a set of four objects or diagrams: (a) a small sufuria or circular tin lid, (b) a bottle cap (crown cap with 21 teeth), (c) a printed diagram of a 6-tooth gear wheel, and (d) a printed top-view diagram of the Ngong Hills turbine hub (3 blades). For each object, students: (1) use a pencil or skewer to physically locate and mark the axis of rotational symmetry; (2) slowly rotate the object and tally each time it maps onto itself; (3) record the order of rotational symmetry and the minimum angle of rotation in a table with columns: Object, Axis Description, Order, Minimum Angle. Groups have 12 minutes. The teacher circulates and uses probing questions without giving answers.	Before groups start, say: 'Your job is to observe carefully and record evidence — not to guess. Use the definition from Lesson 6: the object maps onto itself n times, so the order is n. For 3D objects, describe the axis in words, for example vertical through the centre, or horizontal through the hub.' Circulate and use these targeted questions: For the sufuria — 'Does it map onto itself after 1 degree of rotation? After 10 degrees? What does that tell you about the order?' For the bottle cap — 'Count the teeth carefully. How does the number of teeth relate to the order?' For the gear — 'Compare this to the regular hexagon from Lesson 6. What is the connection?' For the turbine diagram — 'We already found the order in Lesson 6. What is new about describing this as a 3D object?' WAIT TIME: after each probing question, wait at least 10 seconds before adding any further hint. If a group is stuck on the sufuria being order infinity, ask: 'Is	Hands-on investigation with structured recording — the table format requires students to use precise language for each column. The progression from a familiar cooking object to an engineering diagram mirrors the unit storyline from everyday life to the turbine.	Walk the room and check tables after 8 minutes. Listen for: correct axis descriptions (vertical, through the centre of the base); correct orders (sufuria with no handle approximately infinite or a large number if students treat it as a mathematical cylinder — accept this and plan to address it; bottle cap order 21; gear order 6; turbine order 3). Flag groups confusing axis of symmetry with line of reflection symmetry for targeted correction in the Explain phase.

		a perfect mathematical circle the same as a real sufuria with a handle or a spout? Look again.' Note which groups distinguish 2D order from 3D axis confidently.		
Explain Phase 📹 **VIDEO:** Electrochemical gradients and secondary active transport Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Even and Odd Functions (Symmetry) (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. One group presents their table for the bottle cap and gear, another for the sufuria and turbine. After each presentation, students may challenge or add. The teacher then formalises the distinction: for 2D figures, rotational symmetry is described by order and minimum angle; for 3D objects, we add the axis of rotational symmetry — the fixed line about which rotation occurs. Students copy a summary definition into their exercise books. They then complete two consolidation questions individually: (Q1) A Kenyan 20-shilling coin is circular with a coat-of-arms design. Describe its axis of rotational symmetry and estimate its order. (Q2) A regular hexagonal Maasai beadwork pendant is mounted on a flat card. State its order of rotational symmetry and minimum angle of rotation, and explain why we describe an axis rather than a line when the pendant is picked up and held in space.	During presentations, use: 'Can anyone add evidence to support or challenge that claim?' After both presentations, say: 'I am going to give you the formal mathematical definition now. Listen for what is new compared to Lesson 6.' Write on the board: AXIS OF ROTATIONAL SYMMETRY — the fixed straight line about which a 3D object is rotated so that it maps exactly onto itself. ORDER OF ROTATIONAL SYMMETRY — the number of times the object maps onto itself in one complete rotation of 360 degrees. Minimum angle equals 360 degrees divided by order. Say: 'Notice that order works the same way in both 2D and 3D. The new idea today is the axis — a line in 3D space, not just a point on a flat page. In Lesson 5 we found the centre of rotation as a point; today that point extends into a line when we go into 3D.' For the consolidation questions, allow 5 minutes of silent individual work. WAIT TIME: after issuing Q1 and Q2, say nothing for 3 minutes. Circulate and read responses. Cold-call one student for Q1 and a different student for Q2. For Q2, listen specifically for whether the student distinguishes axis (3D, a line) from centre (2D, a point).	Explicit contrast between 2D and 3D language — the teacher draws a T-chart on the board: 2D Figure on one side, 3D Object on the other, listing: centre of rotation vs axis of rotational symmetry; order (same definition); minimum angle (same formula). This visual anchor helps students hold both concepts together.	Collect the written responses to Q1 and Q2 for a quick review after class. During cold-calling, assess whether students: (1) correctly identify the axis as vertical through the centre of the coin; (2) articulate that order for a plain circle is theoretically infinite but a real coin with a design has a finite order; (3) correctly distinguish axis from point for the 3D pendant scenario. Use errors in Q2 to plan a brief 3-minute correction opener for Lesson 8.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Electrochemical	Students return to the class Driving Question Board. The teacher reads aloud the current best class answer to the driving question: 'The Ngong Hills turbine blades always look the same	Say: 'Our driving question asks why the blades always look the same and how we can predict where any point lands. In Lesson 6 we answered the why using order of symmetry. Today we are making	Cumulative model refinement — the DQB serves as the class running theory of the phenomenon. Each lesson deposits new evidence. Students see their understanding becoming	Read sticky notes to assess whether students are correctly using axis language and connecting it to the 3D reality of the turbine. If most notes still say only order 3 without mentioning

gradients and secondary active transport Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Even and Odd Functions (Symmetry) (PLIX) Source: CK-12 🔍 Search ARES for similar readings	because the assembly has rotational symmetry of order 3 — it maps onto itself at 120 degrees, 240 degrees, and 360 degrees.' The teacher then asks: 'What new, more precise language can we now add to this answer?' Students individually write one sentence on a sticky note that adds either the axis or the 3D dimension to the answer. Sticky notes are added to the DQB under the label NEW EVIDENCE — LESSON 7. The class reads three or four notes aloud and the teacher annotates the board: draws an arrow from the turbine diagram to a label reading AXIS: horizontal shaft through the hub, perpendicular to the plane of the blades.	that answer more complete — the turbine is not a flat drawing, it is a real machine spinning in 3D space. Your sticky note should upgrade our answer with that 3D precision.' After students post their notes, read three aloud and ask the class: 'Is this note adding something genuinely new, or is it repeating what we already knew?' WAIT TIME: 8 seconds after asking this question before taking responses. Annotate the turbine diagram on the DQB with the axis label. Point to remaining unanswered DQB questions: 'Next lesson we will tackle: are the three blades actually congruent to each other? What mathematical proof do we need?'	more complete and more precise lesson by lesson, motivating the remaining lessons.	axis or 3D, the teacher knows the 2D-to-3D extension has not yet transferred and must be re-addressed at the start of Lesson 8.
Model Building Phase 📹 **VIDEO:** Electrochemical gradients and secondary active transport Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Even and Odd Functions (Symmetry) (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In the final 6 minutes, each student updates their personal cumulative rotation model — a diagram they have been building since Lesson 1. Today they must add: (1) a labelled 3D sketch of the turbine showing the axis of rotational symmetry; (2) the order and minimum angle for the turbine; (3) one other Kenyan object from today's lesson with its axis and order labelled. Below the sketches, students write a two-sentence updated answer to the driving question that includes the words axis, order, and 3D. Students who finish early add a second real-world Kenyan object of their own choice.	Say: 'Your model is your personal scientific record of everything we have discovered. A geologist does not just remember rock layers — they draw and label them. You are doing the same for rotation. Make sure a student who missed today could read your model and learn both the new idea and how it connects to the turbine.' Circulate and look for models that lack the axis label or that draw the axis incorrectly as a point. Quietly say to those students: 'An axis is a line, not a dot — show me where that line runs through the turbine in 3D.' In the last 2 minutes, ask two volunteers to read their two-sentence driving-question answer aloud. WAIT TIME: after the first volunteer reads, wait 5 seconds before asking the class, 'Does this answer include both order and axis? Thumbs up or thumbs down.' Use thumbs-down responses to prompt a peer to suggest an improvement rather than	Personal model as cumulative scientific record — each student maintains ownership of their developing understanding. The two-sentence written answer forces students to synthesise rather than simply label diagrams, pushing for conceptual integration of 2D order and 3D axis.	Scan models for: correct 3D sketch with axis as a line through the hub; correct order (3) and minimum angle (120 degrees) labelled; at least one additional object with correct axis description. The two-sentence answers reveal whether students have achieved the SLO of distinguishing and integrating order and axis language. Use these as evidence for planning the Lesson 8 opener.

		correcting it yourself.		

D. TEACHER REFLECTION
(1) Did students successfully move from the 2D concept of order of rotational symmetry to the 3D concept of axis of rotational symmetry, or did most remain anchored in 2D language only? (2) Which objects in the Observe phase generated the richest discussion — the sufuria, the bottle cap, the gear, or the turbine — and what does this reveal about which contexts are most productive for this class? (3) Were the WAIT TIME pauses long enough to allow all learners, including those who process more slowly, to formulate responses before peers answered? (4) Do the sticky notes and personal models show that students can now give a more complete answer to the driving question than after Lesson 6? What gaps remain before Lesson 8 on congruence?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that 3D objects such as a sufuria, a bottle cap, a gear, and the Ngong Hills turbine hub each rotate about a fixed line called an axis of rotational symmetry, and that they map onto themselves a countable number of times in one full rotation.
What did I learn?	The axis of rotational symmetry is the fixed straight line about which a 3D object maps onto itself during rotation. The order of rotational symmetry and minimum angle formula (360 degrees divided by order) apply in exactly the same way in 3D as in 2D. The turbine rotates about the horizontal shaft through its hub — this is its axis — and has order 3 with minimum angle 120 degrees.
How does this explain the phenomenon?	The reason the Ngong Hills turbine blades always look the same can now be described with full 3D precision: the assembly has rotational symmetry of order 3 about the horizontal axis through the hub, meaning every 120 degrees of rotation maps the blade assembly exactly onto itself. This is not just a flat geometric property — it is a physical property of the real machine spinning in three-dimensional space.

LESSON 8 (80 minutes): Congruence and Rotation: Proving the Turbine Blades Are Congruent	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To formally establish that rotation is an isometric (congruence-preserving) transformation, and to use this property to prove that the three blades of the Ngong Hills wind turbine are congruent to one another.
Knowledge	Students will know that rotation preserves length, angle size, and area (isometry); that rotation is a congruence transformation alongside reflection and translation; and that two figures are congruent if one can be mapped onto the other by a rotation, reflection, or translation.
Skills	Students will be able to: (a) verify that corresponding side lengths and angles are equal after a rotation; (b) compare rotation, reflection, and translation as congruence transformations; (c) construct a geometric argument — using rotation — that proves two figures are congruent; (d) apply isometry properties to the turbine-blade context.
Attitudes	Students will appreciate the power of mathematical proof as a tool for explaining real-world phenomena, and develop confidence in constructing and presenting evidence-based geometric arguments.
Key Inquiry Question	1. What are the properties of rotation as a geometric transformation? 2. How do we determine the centre and angle of rotation given an object and its image?
Purpose in Storyline	Lesson 8 synthesises the measurement and construction skills built in Lessons 1–7 into a formal proof. Students now have enough evidence — rotation rules, angle measurement, coordinate images — to answer a core part of the driving question: the turbine blades always look the same because each blade is the image of another under a 120° rotation, and rotation preserves shape and size exactly. This lesson closes the loop between congruence and the visual symmetry students first noticed in the phenomenon video.
Safety Notes	No physical hazards. Ensure compasses (for angle/arc drawing) are handled carefully; students should not leave compass points uncapped on desks. Remind students to handle rulers and protractors with care.

B. LESSON OVERVIEW
Students revisit the Ngong Hills turbine phenomenon and pose the question: are the three blades actually congruent? Through a structured predict–observe–explain sequence, they first predict whether rotation changes a shape's size or angles, then measure pre-drawn rotation images on worksheets to gather evidence, and finally formalise the isometry properties of rotation. A short video consolidates the comparison between rotation, reflection, and translation as congruence transformations. Students construct a written geometric argument proving that Blade 2 is the image of Blade 1 under a 120° rotation about the hub, and therefore the blades are congruent. The DQB is updated and cumulative models are revised to include the isometry label.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Properties of Equality and]	Students are shown two screenshots side-by-side on the screen: (A) a single turbine blade from the Ngong Hills turbine in its	Display the two blade images prominently. Say: 'Before we do any measuring today, I want you to commit to a prediction — write it	Prediction Commitment — students commit a written prediction before instruction, activating prior intuitions about	Observable indicator: Every student has a written prediction with a stated reason in their book before the phase ends. Teacher

Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	original position, and (B) the same blade after a 120° rotation about the central hub. The teacher asks: 'Look at these two blade outlines. Do you think they are exactly the same shape and size — or has the rotation stretched, squashed, or changed something about the blade?' Students write a prediction in their exercise books under the heading: 'Does rotation change the shape or size of an object? Give a reason for your prediction.' Students then share predictions with a shoulder partner for 2 minutes before selected individuals share with the class.	down so we can check it later.' Allow 3 minutes of individual silent writing, then 2 minutes of partner discussion. Use cold-call: select 4 students (mix of confidence levels) to share predictions. Record each prediction on the board verbatim under two columns: 'Rotation DOES change shape/size' vs 'Rotation does NOT change shape/size'. Say: 'We are going to gather evidence to find out who is right. Keep your prediction — we will return to it.' Apply 10–12 seconds of wait time after each cold-call before probing with: 'Can you say more about WHY you think that?'	transformation and creating cognitive dissonance that motivates the evidence-gathering phase. Recording predictions publicly on the board creates accountability and a reference point for revision.	scans the room to confirm. Note the proportion of students predicting 'no change' vs 'change' — this reveals baseline understanding of isometry and guides the depth of emphasis needed in the Explain phase.
Observe Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive a structured worksheet containing three pre-drawn rotation tasks set in the turbine context: (1) A simple quadrilateral blade cross-section labelled PQRS rotated 90° anticlockwise about point O — students measure all four sides and all four angles of both the object and image using rulers and protractors, recording in a comparison table. (2) A triangle representing a blade tip, rotated 120° anticlockwise about the turbine hub — same measurement task. (3) A short reading extract (3 sentences): 'Dr. Wanjiru Mwangi, a Kenyan mechanical engineer, designed the blade assembly for a wind farm near Kericho. She noted that each blade must be congruent to the others, otherwise vibrations would destroy the turbine. She relied on the mathematical property that rotation maps one blade perfectly onto another.' After measuring, students complete the comparison table and answer: 'What do you notice about the side	Distribute worksheets. Say: 'Your job right now is to be measurement detectives — use your ruler and protractor carefully and record every measurement. Do not skip any cells in the table.' Circulate and check measurement technique — specifically watch for students measuring the image's sides with the protractor instead of the ruler. Prompt: 'Which tool measures length? Which measures angle?' After 12 minutes of individual measurement, pause the class: 'Hold your pencils. Look at your comparison table. What pattern are you beginning to see?' Wait 12 seconds. Cold-call 3 students. Then play the video (4 minutes). After the video, say: 'Turn back to your prediction. Put a tick or a cross next to it — does the evidence support or challenge what you predicted?' Allow 2 minutes. Cold-call 2 students to narrate what they predicted and what the evidence showed.	Data Table Analysis — students generate their own quantitative evidence by measuring object and image, then compare corresponding values systematically. This moves understanding from visual intuition ('it looks the same') to mathematical evidence ('the measurements prove it'). The reading extract anchors the mathematics in a real Kenyan engineering context, reinforcing why isometry matters.	Observable indicator: Students' comparison tables show correct measurements with object and image values that match (within ±1 mm and ±1°). Teacher checks 6–8 tables during circulation. Red flag: any student whose image measurements differ significantly from object measurements — this signals a measurement error or a misidentification of the image vertices, which must be corrected before the Explain phase.

	lengths and angles before and after rotation?' A short 4-minute video clip (teacher narrated or digital) shows animated rotation of a Kenyan flag motif — a Maasai shield — being rotated 90° and 180°, with length and angle measurements displayed on screen to confirm they are preserved.			
Explain Phase 📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class discussion to formalise the isometry properties. Students copy the following definition into their notes: 'A transformation is an isometry if it preserves distance (length), angle size, and area. The image is congruent to the original object.' Students then work through three teacher-led worked examples on the board: Example 1 — Given triangle ABC rotated 90° anticlockwise about the origin to give triangle A'B'C', show that AB = A'B', BC = B'C', CA = C'A', and angle BAC = angle B'A'C'. Example 2 — A parallelogram representing a section of a sukuma wiki farm plot is rotated 180° about its centre — students verify congruence using side lengths from coordinates. Example 3 — The key turbine example: Blade 1 is a triangle with vertices at hub H(0,0), tip T_1(0, 5), and edge E_1(1, 3). Under a 120° anticlockwise rotation about H, find T_2 and E_2 (teacher provides the rotated coordinates), then students calculate HT_1, HT_2, HE_1, HE_2 and the angle at H — confirming they are equal. After worked examples, students complete a comparison table: Rotation vs Reflection vs Translation — which properties (length, angle, area, orientation) are preserved by each? Students	Write on the board: 'ISOMETRY = same measure'. Say: 'The word comes from Greek — iso means same, metry means measure. A rotation is an isometry because every measurement stays the same.' Work through Example 1 step by step, narrating: 'I measure AB on the object — I get 5 units. Now I measure A'B' on the image — I also get 5 units. This is not a coincidence. It is a mathematical property of rotation.' For Example 3, say: 'This is the turbine. We are now proving — not just guessing — that the blades are congruent.' After the comparison table pair task, cold-call 4 pairs (one per property row). Ask probing questions: 'Does reflection preserve orientation? Think about looking in a mirror — does your left hand become your right hand?' Wait 10 seconds. 'Does rotation preserve orientation? Think about the turbine blade — does it flip over as it rotates?' Wait 10 seconds. Emphasise: 'Rotation and translation preserve orientation. Reflection reverses it. All three preserve length, angle, and area — all three are congruence transformations.'	Worked Example Analysis with Generalisation — teacher models three progressively complex examples (abstract polygon → real-world farm plot → turbine blade), then students extract the general property. The comparison table forces students to place rotation within the broader family of congruence transformations, building relational understanding rather than isolated facts.	Observable indicator: Students correctly complete the comparison table, marking rotation, reflection, and translation each as preserving length, angle, and area, and correctly distinguishing that only reflection reverses orientation. Teacher poses an exit check oral question: 'If I rotate a shape, is the image congruent to the original?' — expects a unanimous 'Yes, because rotation is an isometry.' Any student answering 'No' or 'Sometimes' is flagged for small-group follow-up.

	discuss in pairs for 3 minutes and present findings.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board. The teacher reads aloud the driving question: 'Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate — and how can we use the mathematics of rotation to predict exactly where any point on the blade will land?' Students write a new sticky note (or DQB entry) responding to the prompt: 'Today I can explain part of the driving question. The three turbine blades look the same because…' Students post their notes. The teacher also adds a class-level summary card to the DQB: 'Rotation is an isometry — it preserves length, angle, and area. Each blade is the image of the previous blade under a 120° rotation. Therefore the blades are congruent.' Students identify which questions on the DQB from earlier lessons are now fully answered (e.g., 'Does the blade change shape as it spins?') and which questions still remain open (e.g., 'How do we calculate exactly where a point on the blade lands using coordinates?').	Say: 'We have been building our understanding since Lesson 1. Look at the questions we posted earlier. Which ones can we now answer with confidence?' Give students 2 minutes to scan the DQB silently. Cold-call 3 students to identify a question they believe is now answered and to state the answer. Say: 'Notice how the question about the blades looking the same is now fully answered by mathematics — not just by looking at a photo. That is the power of proof.' Point to remaining open questions: 'These are what we still need to resolve in Lessons 9 and 10.' Add the class summary card to the DQB visibly.	Driving Question Board Synthesis — students articulate their growing understanding in their own words by linking today's isometry finding directly back to the phenomenon. This consolidates the evidence-to-explanation arc and makes the progress of the unit visible to all learners.	Observable indicator: Student DQB sticky notes correctly use the word 'congruent' or 'same shape and size' and reference rotation as the reason. Teacher reads 6–8 notes before the next lesson to identify any persistent misconceptions (e.g., students who write 'the blades look the same because they are painted the same colour' — indicating they have not connected the geometric argument to the visual phenomenon).
Model Building Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos	Students retrieve their cumulative rotation model (started in Lesson 1 and revised each lesson). Today they add a new layer: they draw or annotate a diagram of the three-bladed turbine in which all three blades are labelled, the 120° rotation angles between them are marked, and the following annotation is written alongside: 'Blade 2 is the image of Blade 1 under a 120° anticlockwise rotation	Say: 'Your model should now tell a complete story about why the turbine blades are congruent. Add the isometry annotation today — this is the mathematical proof that explains what your eyes saw in the very first video we watched.' Circulate and check that students are correctly labelling the 120° angles between blades (not the full 360° or an incorrect subdivision). Prompt students who draw only	Cumulative Model Revision — students physically add the isometry layer to a model that has grown across 8 lessons, making their conceptual development visible and tangible. The before-and-after reflection ('The biggest change in my model is…') promotes metacognitive awareness of learning progress.	Observable indicator: The student's model diagram correctly shows three blades separated by 120° angles, labelled as congruent, with a written isometry annotation. The formal proof statement uses the words 'preserves distance and angle measure' and 'congruent'. Teacher collects 5 models at random to review before Lesson 9 as a between-lesson formative check.

📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	about hub H. Blade 3 is the image of Blade 2 under a further 120° anticlockwise rotation. Since rotation is an isometry, all three blades are congruent — same length, same angles, same area.' Students also write a formal two-sentence proof statement: 'Rotation is an isometry because it preserves distance and angle measure. Therefore, any figure and its rotational image are congruent.' Students compare their current model with their Lesson 1 model and write one sentence: 'The biggest change in my model between Lesson 1 and today is…'	two blades: 'If Blade 1 rotates 120° to give Blade 2, what single rotation takes Blade 1 to Blade 3?' Wait 12 seconds. Expected answer: 240° anticlockwise (or equivalently 120° clockwise). At the close of the phase, ask one student to hold up their model and explain it to the class in 30 seconds — a 'model gallery moment'. Conclude: 'In Lessons 9 and 10 we will use coordinates to predict exactly where any point on the blade lands — making our model even more precise.'		

D. TEACHER REFLECTION

1. Did students successfully connect the isometry property back to the turbine phenomenon, or did the formal mathematics feel disconnected from the video they saw in Lesson 1? If disconnected, plan a 5-minute re-anchoring opener for Lesson 9 that replays the turbine video and asks students to narrate it using their new vocabulary (isometry, congruent, 120°). 2. How many students correctly completed the comparison table (rotation vs reflection vs translation)? If fewer than 70% got the orientation row correct, dedicate the first 8 minutes of Lesson 9 to a mirror activity that dramatises orientation reversal in reflection. 3. Were students able to articulate the formal proof statement in their own words, or did they copy it verbatim without understanding? Check DQB sticky notes and model annotations for evidence of genuine paraphrase. 4. Which students are still struggling with the 120° blade spacing? These students likely have a gap in fraction-of-turn understanding from Lesson 2 — consider a brief peer-tutoring pairing at the start of Lesson 9. 5. How confidently did students use the term 'congruent' — is it becoming part of their working vocabulary or still feels foreign? Note frequency of spontaneous student use during pair and cold-call talk.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed (through direct measurement on worksheets) that when a shape is rotated, the side lengths and angle sizes of the image are identical to those of the original object. A video of a Maasai shield motif being rotated also showed on-screen measurements remaining unchanged after rotation.
What did I learn?	Students learned that rotation is an isometry — a transformation that preserves length, angle size, and area — making the image congruent to the original. They learned that rotation, reflection, and translation are all congruence transformations, but only reflection reverses orientation. They formalised that the three Ngong Hills turbine blades are congruent because each is the image of the previous blade under a 120° anticlockwise rotation about the central hub.
How does this explain the phenomenon?	Students explained that the turbine blades always look the same because rotation is an isometry: no matter how far the blades spin, their shape and size cannot change. Each blade is mapped perfectly onto the next by a 120° rotation, proving the three blades are congruent — which is exactly why the turbine appears visually identical in every rotational position.

LESSON 9 (80 minutes): Applications of Rotation in Real-Life Situations	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to apply the properties and mathematics of rotation to solve real-life problems in Kenyan contexts.
Knowledge	Learners will know how rotation is used in real-life Kenyan contexts including logo design (rotational symmetry), clock hand angle calculations, Earth's rotation creating day and night, and gear ratios in machinery.
Skills	Learners will be able to: (a) calculate the angle swept by a rotating object in a given time; (b) determine the order of rotational symmetry of a design; (c) apply rotation properties to solve problems involving gears and Earth's rotation; (d) present a real-life rotation problem and its mathematical solution.
Attitudes	Learners will appreciate that the mathematics of rotation is embedded in everyday Kenyan life — from the Ngong Hills wind turbines to clock faces, the Earth's daily spin, and engineered machinery — and take pride in connecting classroom mathematics to their environment.
Key Inquiry Question	How do we determine the centre and angle of rotation given an object and its image?
Purpose in Storyline	This is Lesson 9 of 10 in the evidence-gathering sequence. Students have already established the properties of rotation, practised constructions, explored rotational symmetry, and calculated angles. In this lesson, they consolidate all prior evidence by applying rotation mathematics to authentic Kenyan real-life contexts — designing rotationally symmetric logos, calculating clock-hand and Earth-rotation angles, and analysing gear ratios. This advances the driving question by demonstrating that the Ngong Hills turbine blades' 'sameness' is not unique: the same mathematics governs clocks, the Earth, and any rotating machine.
Safety Notes	No physical hazards anticipated. Ensure group presentations are time-managed to avoid learner anxiety. Remind learners to handle rulers, protractors, and compasses carefully during any sketching activity.

B. LESSON OVERVIEW

Learners revisit the Ngong Hills wind turbine phenomenon and connect it to a broader question: where else does rotation mathematics appear in Kenyan life? Working in groups, they rotate through four real-life application stations — (1) designing a rotationally symmetric Kenyan logo, (2) calculating the angle swept by clock hands, (3) modelling Earth's rotation and the creation of day and night, and (4) exploring gear ratios in machinery. Each group then prepares and delivers a short presentation of one real-life rotation problem and its mathematical solution, explicitly linking back to the turbine driving question. The lesson serves as an evidence consolidation and public sensemaking event before the final Lesson 10 assessment.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Another Pythagorean theorem proof]	Learners are shown a still image of the Ngong Hills wind turbine alongside four other images: a Kenyan shilling coin (showing a rotationally symmetric design), an	Display the four images on the board or projected screen simultaneously. Say: 'Before we do anything else today, I want you to look at these four images	Activation of Prior Knowledge via Prediction Journaling — By committing predictions to writing before instruction, learners surface their existing mental models about	Observable indicator: Every learner's book contains four written predictions (centre, angle, connection to turbine) before group discussion begins. Teacher

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	analogue clock face, a globe of the Earth tilted on its axis, and a bicycle gear cog. Individually, learners write responses to the prompt: 'Each of these four objects involves rotation. For each one, predict: (a) What is the centre of rotation? (b) What angle or fraction of a full turn does the rotation involve? (c) How does this connect to what we already know about the wind turbine?' Learners record predictions in their exercise books before any discussion.	alongside our turbine. In your book, write your predictions for each — centre of rotation, angle or fraction of a full turn, and how it connects to the turbine. You have 4 minutes. Work silently and individually.' Set a visible timer for 4 minutes. Circulate, reading over shoulders — do NOT correct at this stage. After 4 minutes, say: 'Now turn to your neighbour and compare your predictions for just one minute.' After the pair-share, cold-call 4 learners (one per image) using the class register: 'Amina, tell me your prediction for the clock. Ochieng, what about the Earth? Wanjiru, the coin? Kipchoge, the gear?' After each response, allow 10 seconds of WAIT TIME before probing: 'Does anyone predict differently? Why?' Do not confirm or correct — tell the class: 'We will test all of these predictions today.'	rotation (built across Lessons 1–8). This creates cognitive tension when evidence contradicts predictions, making the subsequent sensemaking more durable.	scans for common misconceptions — e.g., learners who write 'no centre of rotation' for the Earth (confusing axis with a point), or who cannot estimate clock-hand angles — and notes names for targeted questioning during station work.
Observe Phase 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners rotate through four evidence-gathering stations in groups of 4–5 (approximately 10 minutes per station, teacher signals rotation with a clap or bell). Station 1 — Logo Design (Rotational Symmetry): Learners are given a printed outline of a partial logo (a single 'blade' shape inspired by the Kenya Wildlife Service or a local county logo) and a protractor/compass. They must rotate the blade about a marked centre to complete a logo with rotational symmetry of order 3 (120° rotations) and order 4 (90° rotations), recording the angle and direction (positive = anticlockwise). Station 2 — Clock Hand Angles: Learners receive a worksheet showing six analogue clock faces (e.g., 3:00, 6:00, 12:15, 9:30, 2:45,	Before stations begin, spend 3 minutes briefing the class: 'Each station has written instructions — read them carefully before you start. Your job is to gather evidence that helps you answer the driving question: why does the turbine always look the same, and how do we predict where any point lands?' Assign groups and stations. During rotations, spend approximately 2–3 minutes at each station. At Station 2, if learners are struggling with clock angles, prompt: 'The minute hand makes a full 360° in 60 minutes. So how many degrees per minute?' — then allow 12 seconds of WAIT TIME before offering: 'That means 6 degrees per minute.' At Station 3, ask: 'What is the centre of rotation for the Earth? Is it a point	Carousel Evidence Gathering — Rotating stations allow all groups to encounter the same four bodies of evidence, ensuring a shared class knowledge base before the presentation phase. Each station is designed as a mini-investigation that mirrors the structure of the main phenomenon (identify centre, measure angle, predict image position), reinforcing the rotation framework built across Lessons 1–8.	Observable indicators: (1) At Station 1, learners correctly identify 120° and 90° as the rotation angles for order-3 and order-4 symmetry, and mark the direction as positive (anticlockwise) or negative (clockwise). (2) At Station 2, learners correctly calculate that the minute hand sweeps 6° per minute. (3) At Station 3, learners state 15°/hour and correctly calculate 180° for the Nairobi midday-to-midnight rotation. (4) At Station 4, learners correctly determine that Gear B rotates 270° when Gear A rotates 90° (ratio 24:8 = 3:1). Teacher notes which groups are struggling with gear ratio and redirects with the teeth-counting scaffold.

	7:20) and must calculate: (a) the angle swept by the minute hand, (b) the angle swept by the hour hand, and (c) whether the rotation is positive (anticlockwise) or negative (clockwise) using the sign convention established in earlier lessons. Station 3 — Earth's Rotation: Learners read a short illustrated fact card: 'The Earth rotates on its own axis once every 24 hours. At any moment, one side faces the Sun (day) and the other faces away (night). Nairobi is at approximately 1° S latitude.' They answer: (a) How many degrees does the Earth rotate in 1 hour? (b) In 6 hours? (c) If it is midday in Nairobi, what rotation angle must the Earth turn before Nairobi faces away from the Sun (midnight)? (d) Sketch the Earth's rotation, labelling the axis (centre of rotation), Nairobi's position, and the angle. Station 4 — Gear Ratios: Learners examine a diagram of two meshed gears: Gear A (driver, 24 teeth) and Gear B (driven, 8 teeth). They answer: (a) If Gear A rotates 90°, how many degrees does Gear B rotate? (b) If Gear A makes one full revolution, how many full revolutions does Gear B make? (c) How does this connect to the turbine — where the central hub (like Gear A) drives the generator (like Gear B)?	on the surface or something else?' — allow 10 seconds WAIT TIME, cold-call 2 learners. At Station 4, ask: 'If the small gear has fewer teeth, will it rotate more or less than the big gear for the same movement?' — allow 12 seconds WAIT TIME. After all rotations, say: 'Each group now has evidence from all four stations. In your groups, spend 5 minutes deciding which station you will present to the class and how you will connect it back to the turbine.'		
Explain Phase 📹 **VIDEO:** [Another Pythagorean theorem proof] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Each group (4–5 minutes per group) presents their chosen real-life rotation problem and solution to the class. Presenters must: (1) State the real-life context and name the centre of rotation. (2) State the angle of rotation and its direction (positive/anticlockwise or negative/clockwise). (3) Show the	Facilitate presentations from the front. After each group presents, say: 'Before questions, does the class agree that they have correctly identified the centre of rotation and the angle? Thumbs up if yes, thumbs sideways if unsure.' Cold-call a 'thumbs sideways' learner: 'Fatuma, what makes you	Gallery Presentation with Peer Critique — Public presentation of problem solutions requires learners to articulate rotation mathematics in their own words and defend their reasoning. The mandatory connection to the turbine phenomenon ensures that new application contexts are	Observable indicators: (1) Each group's presentation correctly names a centre of rotation, states an angle with correct direction sign, and makes an explicit verbal or written link to the turbine. (2) In the consolidation task, learners correctly calculate 270° (45 min × 6°/min), identify it as a negative

📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	calculation or construction. (4) Explicitly connect their context back to the Ngong Hills turbine driving question. After each presentation, the class asks at least one clarifying question. Following all presentations, learners individually complete a consolidation task in their books: 'A Nairobi clock shows 12:00. The minute hand rotates to show 12:45. (a) What angle has the minute hand swept? (b) Is this a positive or negative rotation? (c) If a point at the tip of the minute hand (10 cm from the centre) starts at the 12 o'clock position, describe where it will be after this rotation.' Learners also revisit and annotate their original predictions from Phase 1 — writing in a different colour: 'I was right / I need to revise this because…'	unsure?' Allow 10 seconds WAIT TIME. After all presentations, project or write the consolidation task on the board. Say: 'You have 8 minutes individually. This connects everything — clock hands, angles, direction, and the position of a point. Go.' Circulate and target learners you identified in Phase 1 as holding misconceptions. After 8 minutes, cold-call 3 learners to share answers: 'Mwangi, what angle did you get? Aisha, is it positive or negative and why? Otieno, describe where the tip of the hand is.' After each answer, ask the class: 'Do you agree?' Allow 8 seconds WAIT TIME for class consensus. Conclude the Explain phase by saying: 'Every single one of these situations — the clock, the Earth, the gears, the logo — obeys exactly the same rules we discovered from the turbine. Centre of rotation. Angle. Direction. Image position. The mathematics is the same.'	integrated into the existing explanatory model rather than stored as isolated facts. The prediction annotation (revising in a different colour) makes conceptual change visible and metacognitive.	(clockwise) rotation, and describe the tip of the minute hand as being at the 9 o'clock position. (3) Prediction annotations show evidence of revised thinking. Teacher collects books at the end of the lesson to check consolidation task responses and identify learners needing support before Lesson 10.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Another Pythagorean theorem proof] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometric Functions and Angles of Rotation (Flexbook)]	The class revisits the Driving Question Board displayed on the classroom wall. Learners are given sticky notes in two colours: GREEN for 'Evidence we now have' and ORANGE for 'Questions still remaining.' Each learner writes at least one green note and one orange note, then posts them on the DQB under the relevant columns. As a class, they read the accumulated green notes aloud and the teacher facilitates a brief discussion: 'Look at how much evidence we have gathered across 9 lessons. Which of our original questions can we now fully answer? Which part of the driving question can we now answer most	Hand out sticky notes and say: 'You have 3 minutes. Green note: write one piece of evidence from today or from any lesson so far that helps explain why the turbine blades always look the same. Orange note: write one question you still have — or one thing you are not yet sure about.' After posting, read 4–5 green notes aloud, choosing ones that reference the core ideas: centre of rotation, angle, direction, preserved distances. Then ask: 'Can anyone use today's evidence to answer the first key inquiry question — what are the properties of rotation? Who can give me a complete answer?'	Driving Question Board Evidence Harvest — The DQB functions as a public, cumulative record of the class's growing understanding. Requiring both a green (evidence) and an orange (remaining question) note prevents premature closure and keeps intellectual curiosity alive heading into the final lesson. Reading green notes aloud reinforces the collective knowledge built across the unit.	Observable indicators: Green notes contain specific, accurate mathematical claims (e.g., 'rotation preserves distance from centre,' 'the Earth rotates 15° per hour,' 'positive rotation is anticlockwise'). Orange notes reveal genuine remaining uncertainties (e.g., 'I am not sure how to find the centre of rotation when only given the object and image'). Teacher photographs the DQB to inform Lesson 10 planning and to identify which gaps need explicit addressing in the final lesson.

Source: CK-12 🔍 Search ARES for similar readings	confidently?' Learners also update their personal driving question trackers in their books.	Allow 12 seconds WAIT TIME, cold-call 2 learners. Then say: 'Tomorrow in Lesson 10 we will use everything on this board to fully answer both inquiry questions and the driving question. Make sure you review your notes tonight.'		
Model Building Phase 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative rotation model — the annotated diagram of the Ngong Hills turbine they have been building since Lesson 1. Today, they add a final 'Applications Layer' to their model. On a new section of the model page (or on an attached sheet), learners create a concept map or annotated diagram that connects the turbine hub (centre of rotation) to at least three real-life rotation applications from today's lesson, showing: (a) the context name, (b) the centre of rotation, (c) the angle and direction, and (d) one sentence explaining how the same rotation mathematics governs all of them. Learners label one arrow on their concept map: 'This is why the turbine blades always look the same — the mathematics is universal.'	Say: 'Open your cumulative model from Lesson 1. Today we add the final layer before our assessment lesson. You are going to connect the turbine to at least three real-life applications using a concept map. You have 10 minutes. Be precise — include the centre, the angle, the direction, and one sentence of explanation for each connection.' Circulate and check that learners are using correct mathematical language. Prompt struggling learners: 'What is the centre of rotation for the clock hands? Is the minute hand's sweep positive or negative?' — allow 10 seconds WAIT TIME before scaffolding. With 2 minutes remaining, say: 'Add your final label: This is why the turbine blades always look the same. Write one sentence that uses the words centre of rotation, angle, and preserved distance.' Cold-call 2 learners to read their sentences aloud. Affirm accurate responses and gently correct language errors: 'Almost — remember, we say the distance from the centre is preserved, not the position. Can you revise that?' Close the lesson: 'In Lesson 10 you will use your complete model and all your evidence to answer the driving question fully. Well done today.'	Cumulative Model Revision with Applications Layer — Adding an 'Applications Layer' to the existing model requires learners to integrate new contextual knowledge with established rotation principles rather than treating applications as separate content. The concept map structure makes the universality of rotation mathematics visible and explicit, directly resolving the core puzzle of the phenomenon: the turbine looks the same because rotation mathematics — centre, angle, preserved distance — governs all rotating systems.	Observable indicators: Each learner's concept map correctly links at least three real-life contexts to the turbine, with accurate labels for centre of rotation, angle, and direction in each case. The closing sentence correctly uses the terms 'centre of rotation,' 'angle,' and 'preserved distance.' Teacher uses these models as a pre-assessment of readiness for the Lesson 10 summative task — learners whose models lack precision or connection to the phenomenon are identified for targeted support at the start of Lesson 10.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did learners successfully connect each real-life application back to the Ngong Hills turbine driving question, or did applications feel disconnected from the phenomenon? If disconnected, plan an explicit bridging activity at the start of Lesson 10. (2) Which station generated the most confusion — gear ratios, Earth's rotation, clock angles, or logo design? Revisit that concept briefly at the start of Lesson 10 before the summative task. (3) Were group presentations mathematically accurate, particularly in stating the direction of rotation (positive/anticlockwise vs. negative/clockwise)? Note any persistent sign-convention errors for correction. (4) Review the DQB orange notes and the consolidation task books: what genuine gaps remain? Ensure Lesson 10's assessment allows learners to demonstrate mastery of the specific outcomes they found most challenging. (5) Were all learners actively contributing during station work and presentations, or did stronger learners dominate? Consider strategic grouping adjustments for Lesson 10.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed four real-life Kenyan rotation contexts at carousel stations: a rotationally symmetric logo design, analogue clock faces showing swept angles, an illustrated model of the Earth rotating on its axis to create day and night in Nairobi, and a meshed gear diagram showing the relationship between driver and driven gear rotations.
What did I learn?	Learners learned that: (1) the angle swept by a clock's minute hand is 6° per minute (360° ÷ 60 min), making it a negative (clockwise) rotation; (2) the Earth rotates 15° per hour on its axis, and Nairobi must rotate 180° from midday to midnight; (3) gear rotation angles are inversely proportional to the number of teeth (a 24-tooth gear driving an 8-tooth gear produces a 3:1 rotation ratio); (4) rotationally symmetric logos are produced by rotating a base shape through equal angles (e.g., 120° for order 3, 90° for order 4) about a fixed centre — the same mathematics that governs the Ngong Hills turbine blades.
How does this explain the phenomenon?	Learners explained that all rotating systems — the Ngong Hills turbine, clock hands, the Earth, and gears — are governed by the same rotation mathematics: a fixed centre of rotation, a measurable angle with a direction (positive = anticlockwise, negative = clockwise), and the preservation of distance from the centre. The turbine blades 'always look the same' because each 120° rotation maps the blade assembly perfectly onto itself — a property shared by every rotationally symmetric object, from Kenyan logos to the Earth's daily spin.

LESSON 10 (80 minutes): Unit Consolidation, Model Finalisation & DQB Review — Answering the Driving Question	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all learning from the rotation unit by finalising cumulative models, completing the Driving Question Board, and demonstrating full mathematical understanding of rotation through a summative task set in the context of the Ngong Hills Wind Farm.
Knowledge	Students demonstrate knowledge of: (a) the definition and properties of rotation as a geometric transformation; (b) direction conventions — positive angles indicate anti-clockwise rotation, negative angles indicate clockwise rotation; (c) the relationship between angle of rotation and fractional turns (e.g., 90° = ¼ turn, 120° = ⅓ turn, 180° = ½ turn); (d) how to determine the centre and angle of rotation given an object and its image; (e) order of rotational symmetry and how equally-spaced blades produce it; (f) congruence preservation under rotation — lengths, angles, and areas are invariant.
Skills	Students are able to: (a) perform rotation of points and shapes on the coordinate plane for given centres and angles; (b) determine the angle and centre of rotation from an object–image pair using perpendicular bisectors; (c) calculate the order of rotational symmetry of a figure; (d) construct and annotate a final cumulative rotation model; (e) provide written mathematical justifications that answer the driving question using unit-wide evidence.
Attitudes	Students appreciate the use of rotation in real-life engineering contexts such as wind energy; promote the value of mathematical precision in describing the physical world; demonstrate responsibility by completing and presenting their models accurately; show unity by collaboratively reviewing and closing the Driving Question Board.
Key Inquiry Question	Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate — and how can we use the mathematics of rotation to predict exactly where any point on the blade will land?
Purpose in Storyline	This is the final lesson of the unit. Students have spent nine lessons building understanding — from first observing the turbine phenomenon, to learning rotation notation, direction conventions, centre of rotation, angle measurement, rotational symmetry, and coordinate transformations. Today they synthesise ALL of that evidence into a final model and a final written explanation that fully and mathematically answers the driving question. The DQB is closed: every question students posted in Lesson 1 is answered with a justified mathematical statement. Students also complete a summative task modelled on the turbine scenario, demonstrating mastery.
Safety Notes	No physical hazards in this lesson. Ensure students handle rulers, compasses, and protractors carefully when constructing final models. Remind students that all submitted work must be their own individual effort during the summative task phase.

B. LESSON OVERVIEW

In this 80-minute consolidation lesson, learners return to the Ngong Hills Wind Farm phenomenon that opened the unit and use the full body of evidence collected across Lessons 1–9 to construct a final, annotated rotation model and to deliver a written Final Explanation answering the driving question. The lesson opens with a whole-class DQB Review (Phase 1), where students revisit every question posted in Lesson 1 and provide mathematical answers using unit vocabulary and evidence. Students then work in groups to compare their Lesson 1 naïve model with their current cumulative model, annotating changes and additions (Phase 2 — Model Finalisation). In Phase 3, students individually complete a structured summative task: given a diagram of a turbine blade assembly, they identify the centre of rotation, determine the angle of rotation between two blade positions, state the order of rotational symmetry, verify congruence of the image, and write a justified Final Explanation answering the driving question. The lesson closes with whole-class sharing of Final Explanations and a teacher-facilitated metacognitive reflection on the journey from 'I wonder…' to 'I know, because…'

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are seated in their unit groups. Each group receives their original Lesson 1 DQB sticky notes (questions they posted at the start of the unit) and their Lesson 1 naïve rotation model. The teacher projects the full driving question on the board: 'Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate — and how can we use the mathematics of rotation to predict exactly where any point on the blade will land?' Students are given 5 minutes to silently re-read their Lesson 1 questions and initial model, then individually write a brief response on a fresh sticky note: 'What did you think in Lesson 1 vs. what do you now know?' Students post their 'then vs. now' notes on the DQB alongside their original Lesson 1 questions — creating a visual before/after for the class.	Display the Ngong Hills turbine video clip from Lesson 1 (30 seconds) silently. Say: 'We began this unit watching these blades spin. You had questions. You had guesses. Today we close the loop.' Distribute Lesson 1 DQB sticky notes and naïve models back to each group. Say: 'Take 5 minutes in silence. Read what you wrote on Day 1. Then on a new sticky note, write: What did I think then? What do I know now? Be specific — use the mathematics.' Allow 5 minutes of silent individual writing (WAIT TIME). Cold-call 3 students to share their 'then vs. now' notes before they post them. Use the prompt: 'Tell us one thing you were confused about in Lesson 1 that you can now explain mathematically.' After sharing, say: 'Keep those original questions visible — we will answer every single one of them by the end of this lesson.'	Strategy: 'Then vs. Now' Metacognitive Comparison. Students explicitly compare their prior conceptions with their current understanding. This activates retrieval of all unit learning, surfaces any remaining misconceptions before the summative task, and builds metacognitive awareness of how their thinking has grown. It connects directly to the phenomenon by anchoring the reflection in the same turbine image that opened the unit.	Observable indicator: Each student produces a written 'then vs. now' sticky note that references at least one specific mathematical term from the unit (e.g., centre of rotation, angle, anti-clockwise, order of rotational symmetry, congruence). Teacher scans sticky notes as students post them. Red flag: Any student who writes only descriptive language ('the blades move in a circle') without mathematical language — teacher notes these students for targeted support during the summative task phase.
Observe Phase 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook)	Groups are given 12 minutes to conduct a 'Gallery Walk' around the classroom. Posted on the walls are six evidence stations — one per major concept from the unit — each with a summary prompt card: (1) Definition & Notation of Rotation; (2) Direction Convention — positive = anti-clockwise, negative = clockwise; (3) Angle of Rotation & Fractional Turns (90° = ¼ turn, 120° = ⅓ turn, 180° = ½ turn); (4) Centre of Rotation — how to find it using perpendicular bisectors; (5) Order of Rotational Symmetry — why three equally-spaced blades give order 3; (6)	Set a visible 12-minute timer. Say: 'Each station holds evidence your class generated across this unit. At every station, write exactly one sentence that uses that evidence to help answer: Why do the turbine blades always look the same? You have 2 minutes per station — move on my signal.' Circulate during the gallery walk. At Station 2 (direction convention), pause at any group and ask: 'If a blade rotates negative 120 degrees, which direction does it turn? Show me with your hand.' At Station 5 (order of rotational symmetry), ask: 'How many times does the turbine	Strategy: Evidence Gallery Walk with Targeted Sentence Stems. By physically moving to distinct evidence stations, students re-encounter each sub-concept in a condensed, retrieved form. Writing one sentence per station forces precision and connection to the phenomenon. This strategy builds toward the Final Explanation by giving students six 'building blocks' — one per concept — that they will assemble into a coherent mathematical argument.	Observable indicator: Each group produces six completed sentence cards, each referencing the correct mathematical concept and linking it to the turbine blades. Teacher checks Station 3 cards (angle and fractional turns) specifically — students should write something like: 'A turbine blade rotating 120° anti-clockwise makes a ⅓ turn, mapping the assembly exactly onto itself.' Red flag: Students who write vague sentences ('rotation changes the position') — teacher redirects with: 'What specific property of rotation makes the blade look the same? Use a

Source: CK-12 🔍 Search ARES for similar readings	Congruence Under Rotation — what stays the same (lengths, angles, area). At each station, students write one sentence connecting that evidence to the driving question on a provided card. Groups return to their seats and use their six sentence cards to begin drafting their Final Explanation.	blade assembly map onto itself in one full 360-degree turn?' (Expected answer: 3 times — order 3.) After the gallery walk, cold-call 2 groups to read their sentence from Station 6 (congruence). Listen for the key idea: the image blade is congruent to the object blade because rotation preserves length, angle size, and area.		number or a term from your notes.'
Explain Phase 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students work individually on a structured summative task (printed worksheet). The task presents a coordinate grid showing one turbine blade modelled as a triangle with vertices A(0,0), B(4,0), C(4,2). Students are told the hub (centre of rotation) is at the origin O(0,0). The full turbine has three equally-spaced blades. Tasks: (Q1) State the angle of rotation that maps Blade 1 to Blade 2, giving direction. Justify your answer using the order of rotational symmetry. (Q2) If Blade 1 rotates by –240°, determine the coordinates of the image of point B(4,0) after this rotation. Show all working. (Q3) A second turbine is observed: its blade appears to map from position P(3,0) to image P'(0,3). Determine the angle and direction of rotation about the origin. (Q4) Verify that the image of Blade 1 after rotation is congruent to Blade 1. State two properties that are preserved. (Q5 — Final Explanation, 5 marks): In full sentences, answer the driving question: 'Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate?' Use the following terms: centre of rotation, angle of rotation, order of rotational symmetry, congruence,	Distribute printed summative task worksheets. Say: 'This is your individual summative task. You have 25 minutes. Use your knowledge — not your neighbour's. Show all working for every calculation. For Question 5, write at least 4 sentences and use all five listed mathematical terms.' Set a 25-minute visible timer. For the first 2 minutes, read each question aloud clearly. Remind students: 'Positive rotation angle = anti-clockwise. Negative rotation angle = clockwise. A rotation of 120° is one-third of a full turn.' Circulate silently. Do NOT answer content questions during the summative — redirect with: 'Check your notes and your model.' With 5 minutes remaining, give a time warning: 'Five minutes left — if you have not started Question 5, begin it now.' After 25 minutes, collect worksheets. Cold-call 2 students to share their Q5 Final Explanation aloud before moving to model finalisation.	Strategy: Structured Summative Task with Scaffolded Final Explanation. Questions 1–4 are procedural and verify skills; Question 5 is a constructed-response Final Explanation that requires synthesis. The five mandated terms serve as scaffolding — they ensure students address all five pillars of the unit's conceptual framework. Reading two Q5 responses aloud after collection models high-quality mathematical writing for peers and creates a shared class-level Final Explanation moment.	Observable indicator — Q1: Students write '120° anti-clockwise' (or equivalent: '–240° = 360°–240° = 120° anti-clockwise') with justification referencing order-3 symmetry. Q2: Students correctly apply rotation matrix or known angle rule to find B'(0,4). Q3: Students identify 90° anti-clockwise by inspection or perpendicular bisector reasoning. Q4: Students state that length AB = length A'B' and angle BAC = angle B'A'C' (or equivalent area argument). Q5: Full-sentence response using all five required terms with accurate mathematical content. Teacher marks Q5 using a 5-point rubric: 1pt per term used correctly in context.

	anti-clockwise/clockwise.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups return to the class DQB. Each group is assigned 2–3 of the original Lesson 1 questions from the board. They have 8 minutes to write a mathematical answer to each question on a green sticky note and place it directly beneath the original question, physically 'closing' the question on the board. The class then does a 2-minute silent 'read-around' — everyone walks the DQB and reads all the green answer notes. Students place a tick sticker on any answer they agree with and a question-mark sticker on any they want to challenge. The teacher facilitates a 3-minute whole-class discussion on any questioned answers.	Say: 'In Lesson 1, you posted questions you couldn't answer. Now you can. Each group has been assigned 2 to 3 of those original questions. Write a complete mathematical answer on a green sticky note — include at least one number, formula, or diagram — and post it beneath the question. You have 8 minutes.' Set timer. Circulate and prompt: 'Your answer needs to include a mathematical justification, not just a statement.' After posting, say: 'Two minutes of silence — walk the board, read every green note, place a tick if you agree or a question mark if something seems wrong.' After the read-around, address any question-marked answers: cold-call the group who wrote the challenged answer — 'Defend your answer mathematically in one sentence.' Conclude by pointing to the full DQB and saying: 'Every question posted on Day 1 now has a mathematical answer. That is what it means to understand rotation.'	Strategy: DQB Closure with Peer Validation (Tick & Question-Mark Protocol). Physically closing questions on the DQB creates a tangible sense of intellectual completion. The tick/question-mark protocol introduces low-stakes peer review, requiring students to evaluate mathematical claims — a higher-order thinking move. This strategy reinforces that scientific and mathematical understanding is publicly verifiable, not just individually held.	Observable indicator: Every original Lesson 1 DQB question has a green answer note posted beneath it. Teacher checks that no answer note is purely descriptive — each must contain at least one of: a specific angle value, a direction term, a symmetry order, a coordinate, or a named property. Red flag: Any group that writes 'rotation keeps the shape the same' without specifying which properties (lengths, angles, area) — teacher adds a question-mark sticker and invites class to improve the answer.
Model Building Phase 📹 **VIDEO:** Another Pythagorean theorem proof Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Functions and Angles of	Each student opens their cumulative rotation model (built and revised across Lessons 1–9). Using a fresh annotation colour (red pen), they add final labels to their model: (a) label the centre of rotation O; (b) annotate the angle of rotation between each pair of blades (120° anti-clockwise); (c) write the order of rotational symmetry (Order 3) in a circle at the hub; (d) add a congruence statement: 'Blade 1 ≅ Blade 2 ≅ Blade 3'; (e) add a direction arrow showing anti-clockwise rotation with '+' label and a clockwise arrow	Say: 'Take out your cumulative model. Pick up the red pen. In the next 8 minutes, add the five final annotations I am listing on the board.' Write the five annotations on the board (centre O, 120° anti-clockwise, Order 3, congruence statement, direction arrows). Set an 8-minute timer. Circulate and check: every model must show a numerically labelled angle — not just an arrow. After annotation, say: 'Place your Lesson 1 model next to your final model. Write your Model Evolution Statement at the bottom — two sentences, starting:	Strategy: Model Evolution Statement + Expert Model Comparison. Asking students to articulate 'In Lesson 1 I thought… Now I know… because…' requires them to identify the specific conceptual shift they made, grounding metacognition in mathematical evidence. Comparing their model to an expert model (rather than being told the answer) maintains cognitive engagement and ownership. The five mandatory red-pen annotations ensure the final model is complete and	Observable indicator: Each student's final model displays all five red-pen annotations correctly: (a) centre O labelled, (b) 120° anti-clockwise angle shown between blades, (c) 'Order 3' at hub, (d) congruence statement written, (e) both direction arrows with +/– labels. Model Evolution Statement uses a causal connector ('because') with a mathematical justification. Teacher does a rapid 30-second visual scan of every final model before the lesson ends and marks a checklist: 5/5 annotations present = mastery

Rotation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	with '–' label. Students then place their Lesson 1 naïve model next to their final model and write a two-sentence 'Model Evolution Statement' at the bottom: 'In Lesson 1, I thought… Now I know… because…' Groups briefly share their most improved model with the class (1 minute per group).	In Lesson 1 I thought… Now I know… because…' Cold-call one student from each group to read their Model Evolution Statement. After all groups have shared, display a prepared 'Expert Model' on the projector — a fully annotated teacher version — and say: 'This is what a complete, correct rotation model looks like. Compare yours. What is one thing you would still add or improve?' Allow 2 minutes for final self-correction. Conclude: 'The Ngong Hills turbine blades look the same because rotation is an isometric transformation — it preserves shape and size — and three blades equally spaced at 120° give an order-3 rotational symmetry. You have just proved that with mathematics.'	assessable.	indicator.

D. TEACHER REFLECTION

After this lesson, reflect on the following questions to inform future unit planning and delivery:

1. DRIVING QUESTION RESOLUTION: Did every student produce a Q5 Final Explanation that used all five required mathematical terms (centre of rotation, angle of rotation, order of rotational symmetry, congruence, anti-clockwise/clockwise) in a mathematically accurate sentence? If not, which terms were missing most frequently — and what does that reveal about which lesson in the unit needs strengthening?

2. MODEL QUALITY: When you scanned final models for the five red-pen annotations, what percentage of students had all five? Were there specific annotations (e.g., the direction arrows with +/– labels, or the congruence statement) that were consistently missing? This reveals whether the direction convention (positive = anti-clockwise, negative = clockwise) and congruence properties were adequately consolidated.

3. DQB CLOSURE: Were all original Lesson 1 questions answered with mathematical justification — not just descriptive statements? If groups struggled to write mathematical answers, which questions remained vague? Consider whether the DQB was actively maintained and reviewed throughout the unit or only revisited in this final lesson.

4. SUMMATIVE TASK — Q2 (Coordinate Rotation): Did students correctly rotate B(4,0) by –240° to find B'(0,4)? Common error: students may treat –240° as 240° clockwise rather than recognising it equals 120° anti-clockwise. If more than 30% of students made this error, re-examine how direction conventions were taught in Lessons 3–4.

5. METACOGNITIVE GROWTH: Did students' 'Model Evolution Statements' show genuine conceptual shift, or were they superficial ('I thought blades were just moving; now I know they rotate')? High-quality statements should reference a specific mathematical property — e.g., 'In Lesson 1, I thought the blades changed shape as they moved; now I know rotation is isometric, so the blade is always congruent to its original position, because lengths and angles are preserved under rotation.'

6. PHENOMENON CONNECTION: Throughout the lesson, did students consistently connect their mathematical answers back to the Ngong Hills turbine? Or did the mathematics become abstract and detached from the context? If disconnection occurred, consider opening future units with a more interactive phenomenon experience (e.g., a student-built model turbine or a field visit to Ngong Hills) to deepen the anchor.

7. TIMING: The summative task (25 minutes) and model finalisation (8 minutes) are the most time-sensitive phases. If time ran short, what was cut — and was it a critical component? Consider whether the gallery walk (Phase 2) could be shortened to 8 minutes in future iterations to protect summative task time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students watched the Ngong Hills Wind Farm turbine video again (the same clip from Lesson 1) and observed that the three equally-spaced blades, each separated by 120°, cause the entire blade assembly to map perfectly onto itself three times in one full 360° rotation. They also observed, through the evidence gallery walk, a consolidated display of all unit evidence: direction conventions (positive = anti-clockwise, negative = clockwise), fractional turns (90° = ¼ turn, 120° = ⅓ turn), centre-of-rotation construction, and congruence preservation.
What did I learn?	Students learned that the driving question — 'Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate?' — is fully answered by five interconnected mathematical ideas: (1) Rotation is defined by a centre, an angle, and a direction; (2) A positive angle means anti-clockwise rotation, a negative angle means clockwise rotation; (3) The three blades are equally spaced at 120° intervals, giving the assembly an order-3 rotational symmetry — meaning it maps onto itself 3 times in 360°; (4) Rotation is an isometric transformation, so every blade image is congruent to the original — lengths, angles, and area are all preserved; (5) Given any object–image pair, the centre and angle of rotation can be determined precisely using perpendicular bisectors and angle measurement.
How does this explain the phenomenon?	Students explained the driving question in their own written Final Explanations (Q5 of the summative task) using all five required mathematical terms. The class-level consensus explanation is: 'The blades of the Ngong Hills wind turbine always look the same because the three blades are equally spaced at 120° from one another around the fixed centre of rotation at the hub. When the turbine rotates by 120° anti-clockwise (a positive rotation), the entire blade assembly maps perfectly onto itself — this is order-3 rotational symmetry. Because rotation is an isometric transformation, every blade in its new position is congruent to its original position: all lengths, angles, and areas are preserved. This means no matter how many 120° turns the turbine makes — whether clockwise (negative angles) or anti-clockwise (positive angles) — the shape always appears identical. Using the mathematics of rotation, we can predict exactly where any point on a blade will land after any given angle of rotation about the hub.'

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.